数据准备oom定位及解决

最初现象，使用全局静态类加载器

```
root@pingjia-VMware-Virtual-web-2:/opt/app# jstat -gc 1 1000 5
 S0C    S1C    S0U    S1U      EC       EU       OC        OU       MC     MU    CCSC   CCSU
103424.0 99328.0 40640.7  0.0   1190400.0 608529.6 2796544.0  2508085.8  796876
103424.0 99328.0 40640.7  0.0   1190400.0 1029830.3 2796544.0  2508085.8  79687
96256.0 99328.0  0.0   38151.5 1190400.0 352778.0 2796544.0  2508085.8  797132.(
96256.0 99328.0  0.0   38151.5 1190400.0 870175.5 2796544.0  2508085.8  797132.(
95232.0 92160.0 43175.5  0.0   1199104.0 213486.0 2796544.0  2508085.8  797132.(
```

再次计算，完成之后，老年代持续无法被 FGC 回收，且 FGC 次数从 710 到 765

```
# 使用全局静态类加载器，即使计算完成之后，老年代持续无法被回收。
c1@pingjia-VMware-Virtual-web-2:/opt/app/logs$ jstat -gc 1 2000 5
 S0C    S1C    S0U    S1U      EC       EU       OC        OU       MC     MU    CCSC   CCSU
10752.0 7168.0  0.0   6687.5 727040.0 495621.0 2796544.0  2776146.0  619432.0 37
10752.0 7168.0  0.0   6687.5 727040.0 495621.0 2796544.0  2776146.0  619432.0 37
10752.0 7168.0  0.0   6687.5 727040.0 495623.1 2796544.0  2776146.0  619432.0 37
10752.0 7168.0  0.0   6687.5 727040.0 495625.5 2796544.0  2776146.0  619432.0 37
10752.0 7168.0  0.0   6687.5 727040.0 495723.6 2796544.0  2776146.0  619432.0 37
```

```
c1@pingjia-VMware-Virtual-web-2:/opt/app/logs$ jstat -gc 1 2000 5
 S0C    S1C    S0U    S1U      EC       EU       OC        OU       MC     MU    CCSC   CCSU   YGC     YGCT    FGC    FGCT     GCT
10752.0 7168.0  0.0   6687.5 727040.0 495621.0 2796544.0  2776146.0  619432.0 377417.8 170920.0 62424.6   6339  145.728  765  1539.571 1685.299
10752.0 7168.0  0.0   6687.5 727040.0 495621.0 2796544.0  2776146.0  619432.0 377417.8 170920.0 62424.6   6339  145.728  765  1539.571 1685.299
10752.0 7168.0  0.0   6687.5 727040.0 495623.1 2796544.0  2776146.0  619432.0 377417.8 170920.0 62424.6   6339  145.728  765  1539.571 1685.299
10752.0 7168.0  0.0   6687.5 727040.0 495625.5 2796544.0  2776146.0  619432.0 377417.8 170920.0 62424.6   6339  145.728  765  1539.571 1685.299
10752.0 7168.0  0.0   6687.5 727040.0 495723.6 2796544.0  2776146.0  619432.0 377417.8 170920.0 62424.6   6339  145.728  765  1539.571 1685.299
c1@pingjia-VMware-Virtual-web-2:/opt/app/logs$
```

改为每次都创建类加载器，并主动回收资源；

计算完成后，才最终老年代能被回收，但是计算过程中老年代无法及时回收，导致 FGC 频率很高

FGC 次数多

OC=1398M

OU=225M

MC=446m

MU=185m

```
c1@Bank-AI-Credit-5142:/opt/app$ jstat -gc 1 2000 5
 S0C    S1C    S0U    S1U      EC       EU       OC         OU       MC       MU     CCSC   CCSU
512.0  512.0   0.0   288.0  697856.0 285407.5 1398272.0   225200.0  446208.0 18574
512.0  512.0   0.0   288.0  697856.0 394324.1 1398272.0   225200.0  446208.0 18574
512.0  512.0   0.0   288.0  697856.0 501858.0 1398272.0   225200.0  446208.0 18574
512.0  512.0   0.0   288.0  697856.0 612151.0 1398272.0   225200.0  446208.0 18574
512.0  512.0 320.0    0.0   697856.0 11010.8  1398272.0   225200.0  446208.0 18574
```

```
[devops@Bank-AI-Credit-5142 ~]$ docker exec -it dib-agent-service-data-web /bin/bash
c1@Bank-AI-Credit-5142:/opt/app$ jstat -gc 1 2000 5
 S0C    S1C    S0U    S1U      EC       EU       OC         OU       MC       MU     CCSC   CCSU   YGC     YGCT    FGC    FGCT     GCT
512.0  512.0   0.0   288.0  697856.0 285407.5 1398272.0   225200.0  446208.0 185745.3 110592.0 22844.5  12274  660.678  188   801.700 1462.378
512.0  512.0   0.0   288.0  697856.0 394324.1 1398272.0   225200.0  446208.0 185745.3 110592.0 22844.5  12274  660.678  188   801.700 1462.378
512.0  512.0   0.0   288.0  697856.0 501858.0 1398272.0   225200.0  446208.0 185745.3 110592.0 22844.5  12274  660.678  188   801.700 1462.378
512.0  512.0   0.0   288.0  697856.0 612151.0 1398272.0   225200.0  446208.0 185745.3 110592.0 22844.5  12274  660.678  188   801.700 1462.378
512.0  512.0 320.0    0.0   697856.0 11010.8  1398272.0   225200.0  446208.0 185745.6 110592.0 22844.5  12275  660.688  188   801.700 1462.388
c1@Bank-AI-Credit-5142:/opt/app$
```

统计内存实例，及对应的类 - 静态类加载器

```
10752.0 7168.0  0.0   6687.5 727040.0 495623.1 2796544.0  2776146.0  619432.0 377417.8 170920.0 62424.6   6339  145.728  765  1539.571 1685.
10752.0 7168.0  0.0   6687.5 727040.0 495625.5 2796544.0  2776146.0  619432.0 377417.8 170920.0 62424.6   6339  145.728  765  1539.571 1685.
10752.0 7168.0  0.0   6687.5 727040.0 495723.6 2796544.0  2776146.0  619432.0 377417.8 170920.0 62424.6   6339  145.728  765  1539.571 1685.
c1@pingjia-VMware-Virtual-web-2:/opt/app/logs$ jmap -histo 1 | head -n 10

 num     #instances         #bytes  class name
----------------------------------------------
   1:      10311310      577433360  org.codehaus.groovy.runtime.metaclass.MetaMethodIndex$Entry
   2:       7438022      297520880  java.util.LinkedHashMap$Entry
   3:       7903074      238484400  [Ljava.lang.Object;
   4:       2679399      185828416  [C
   5:       6909731      165833544  org.codehaus.groovy.util.FastArray
   6:       2395117      126704128  [I
   7:       1428219      124098368  [Ljava.util.HashMap$Node;
c1@pingjia-VMware-Virtual-web-2:/opt/app/logs$
```

c1@pingjia-VMware-Virtual-web-2:/opt/app/logs$ jmap -histo 1 | head -n 10

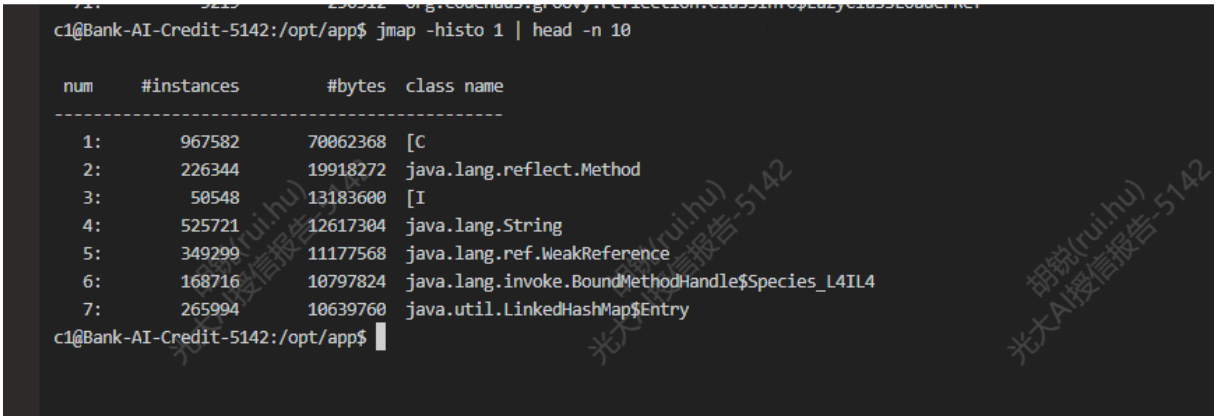
num #instances #bytes class name

1: 10311310 577433360 org.codehaus.groovy.runtime.metaclass.MetaMethodIndex$Entry

2: 7438022 297520880 java.util.LinkedHashMap$Entry

```
   3:      7903074      238484400  [Ljava.lang.Object;
   4:      2679399      185828416  [C
   5:      6909731      165833544  org.codehaus.groovy.util.FastArray
   6:      2395117      126704128  [I
   7:      1428219      124098368  [Ljava.util.HashMap$Node;
c1@pingjia-VMware-Virtual-web-2:/opt/app/logs$
```

统计内存实例，及对应的类 - 每次都创建，并回收类加载器




```
c1@Bank-AI-Credit-5142:/opt/app$ jmap -histo 1 | head -n 10

 num     #instances         #bytes  class name
----------------------------------------------
   1:       967582       70062368  [C
   2:       226344       19918272  java.lang.reflect.Method
   3:        50548       13183600  [I
   4:       525721       12617304  java.lang.String
   5:       349299       11177568  java.lang.ref.WeakReference
   6:       168716       10797824  java.lang.invoke.BoundMethodHandle$Species_L4IL4
   7:       265994       10639760  java.util.LinkedHashMap$Entry
c1@Bank-AI-Credit-5142:/opt/app$
```

todo，
待验证：java 代码中的 groovyshell 的类加载器，是否与脚本中的 new groovyShell 一致？
可以通过 debug 某个脚本看内存地址验证

另外现在要解决的问题是：
为什么指标计算完之后，最终可以回收老年代内存？
但是过程中新生代非常快速就占满，

导致老年代 OU 持续上升，频繁触发 full gc，但是 OU 无法有效的下降
最终定时任务完成后，还是可以回收老年代 OU

打印GC日志，重跑 c1-data 定时任务

```
27136.0 27136.0  0.0   16410.3 644608.0 423244.1 1398272.0  1379438.0  743808.0 434628.9 150400.0 67149.6   6781  354.038  129   531.294  885.332
26624.0 26624.0  0.0   23101.1 645120.0 33833.1  1398272.0  1383638.0  744832.0 435278.4 150656.0 67265.3   6785  354.224  129   531.294  885.518
26624.0 27136.0 22932.2  0.0   644608.0 367854.7 1398272.0  1387084.4  745856.0 435779.6 150912.0 67354.7   6788  354.356  129   531.294  885.650
26624.0 27136.0 22742.7  0.0   644608.0 12853.6  1398272.0  1391236.4  747136.0 436420.6 151168.0 67468.9   6792  354.536  129   531.294  885.830
26624.0 27136.0  0.0   22276.2 644096.0 321269.3 1398272.0  1394372.4  748160.0 436941.7 151424.0 67560.9   6795  354.679  129   531.294  885.973
26624.0 26624.0 22932.2  0.0   645632.0   0.0    1398272.0  1397540.4  748928.0 437433.7 151680.0 67647.8   6798  354.811  130   531.294  886.105
26624.0 26624.0 22932.2  0.0   645632.0   0.0    1398272.0  1397540.4  748928.0 437433.7 151680.0 67647.8   6798  354.811  130   531.294  886.105
26624.0 26624.0 22932.2  0.0   645632.0   0.0    1398272.0  1397540.4  748928.0 437433.7 151680.0 67647.8   6798  354.811  130   531.294  886.105
28672.0 29696.0 9510.7   0.0   639488.0 216573.0 1398272.0  1311107.8  749184.0 437537.2 151680.0 67666.5   6800  354.871  130   539.145  894.016
30208.0 31232.0  0.0   13718.7 636416.0 592988.1 1398272.0  1311107.8  750464.0 438139.6 151936.0 67762.8   6803  354.960  130   539.145  894.105
26624.0 28160.0  0.0   15270.7 642048.0 293521.6 1398272.0  1311107.8  751488.0 438766.9 152192.0 67875.2   6807  355.088  130   539.145  894.233
26112.0 27136.0 20854.7  0.0   644608.0 633739.7 1398272.0  1311107.8  752512.0 439279.3 152448.0 67966.4   6810  355.205  130   539.145  894.350
28672.0 28672.0 20448.0  0.0   641536.0 261059.7 1398272.0  1312658.5  753536.0 439912.1 152704.0 68080.1   6814  355.370  130   539.145  894.515
29184.0 28160.0  0.0   22678.7 641536.0 637632.4 1398272.0  1315850.5  754560.0 440415.3 152960.0 68169.3   6818  355.495  130   539.145  894.640
29184.0 28672.0  0.0   22712.9 640512.0 285940.6 1398272.0  1320074.5  755840.0 441076.3 153216.0 68283.9   6821  355.664  130   539.145  894.809
28160.0 27648.0 17450.4  0.0   643072.0 625846.0 1398272.0  1323194.5  756864.0 441612.4 153472.0 68382.2   6824  355.782  130   539.145  894.927
27648.0 28160.0 22707.7  0.0   642560.0 243086.9 1398272.0  1327394.5  758144.0 442322.9 153728.0 68513.9   6828  355.958  130   539.145  895.103
27648.0 27648.0  0.0   22861.1 643584.0 563631.3 1398272.0  1330554.5  759168.0 442859.5 153984.0 68611.2   6831  356.087  130   539.145  895.232
27136.0 27136.0  0.0   22134.7 644608.0 205078.4 1398272.0  1335013.7  760448.0 443517.1 154240.0 68727.6   6835  356.263  130   539.145  895.408
c1@Bank-AI-Credit-5142:/opt/app$ jstat -gc 1 3000 100
 S0C    S1C    S0U    S1U      EC       EU        OC         OU       MC     MU    CCSC   CCSU   YGC     YGCT    FGC    FGCT     GCT
56320.0 56320.0 38112.0  0.0   586240.0 175803.9 1398272.0  1147285.3  778624.0 431639.2 159104.0 66040.0   6910  359.719  131   542.649  902.368
56320.0 56320.0 30272.0 38112.0 586240.0 586240.0 1398272.0  1151805.3  778624.0 431833.4 159104.0 66078.9   6912  359.776  131   542.649  902.425
56832.0 40960.0  0.0   40819.6 586752.0 434150.8 1398272.0  1154853.3  778624.0 432198.9 159104.0 66150.6   6913  359.902  131   542.649  902.551
55296.0 55808.0  0.0   37875.6 586240.0 246215.9 1398272.0  1159957.3  778624.0 432687.5 159104.0 66253.6   6915  360.020  131   542.649  902.669
53248.0 54272.0  0.0   37619.6 589312.0 82502.2  1398272.0  1165197.3  778624.0 433086.0 159104.0 66332.9   6917  360.141  131   542.649  902.790
53248.0 52736.0 37587.6  0.0   592896.0 521741.4 1398272.0  1167749.3  778624.0 433283.9 159104.0 66372.6   6918  360.202  131   542.649  902.852
```

c1@Bank-AI-Credit-5142:/opt/app$ jstat -gc 1 3000 100

S0C S1C S0U S1U EC EU OC OU MC MU CCSC CCSU YGC YGCT FGC FGCT GCT

524288.0 524288.0 56732.2 0.0 1048576.0 728456.8 2097152.0 70303.2 129484.0 118255.2 17408.0 14625.1 14 0.832 4 0.879 1.711

524288.0 524288.0 0.0 66044.2 1048576.0 973298.0 2097152.0 70311.2 131276.0 119199.0 17920.0 14833.9 15 0.939 4 0.879 1.818

524288.0 524288.0 0.0 84145.3 1048576.0 162792.5 2097152.0 70311.2 134092.0 120840.7 18944.0 15187.7 17 1.211 4 0.879 2.089

524288.0 524288.0 0.0 84145.3 1048576.0 955353.1 2097152.0 70311.2 134092.0 120840.7 18944.0 15187.7 17 1.211 4 0.879 2.089

524288.0 524288.0 0.0 100387.8 1048576.0 20882.8 2097152.0 70359.2 137292.0 122542.2 19840.0 15547.6 19 1.525 4 0.879 2.404

524288.0 524288.0 111071.0 0.0 1048576.0 207947.7 2097152.0 70367.2 139212.0 123496.8 20480.0 15761.3 20 1.699 4 0.879 2.578

524288.0 524288.0 0.0 118995.8 1048576.0 563103.6 2097152.0 70375.2 140364.0 124237.6 20864.0 15911.9 21 1.879 4 0.879 2.758

524288.0 524288.0 128124.5 0.0 1048576.0 752192.3 2097152.0 70375.2 141900.0 125030.9 21376.0 16091.6 22 2.111 4 0.879 2.990

524288.0 524288.0 0.0 133955.8 1048576.0 1019626.6 2097152.0 70415.2 142924.0 125681.0 21632.0 16217.8 23 2.323 4 0.879 3.202

524288.0 524288.0 0.0 131972.7 1048576.0 146340.9 2097152.0 90746.4 145996.0 127400.4 22656.0 16601.6 25 2.751 4 0.879 3.630

524288.0 524288.0 130788.7 0.0 1048576.0 460386.4 2097152.0 100258.4 147532.0 128185.2 23168.0 16764.8 26 2.944 4 0.879 3.823

524288.0 524288.0 0.0 132605.6 1048576.0 774897.9 2097152.0 109602.4 148940.0 128964.8 23552.0 16933.1 27 3.141 4 0.879 4.019

524288.0 524288.0 127794.2 75002.0 1048576.0 1048576.0 2097152.0 125010.4 150476.0 129742.1 24064.0 17104.9 29 3.324 4 0.879 4.203

524288.0 524288.0 130151.5 0.0 1048576.0 398304.3 2097152.0 135413.1 153548.0 131441.2 25088.0 17471.7 30 3.695 4 0.879 4.574

524288.0 524288.0 0.0 129402.0 1048576.0 356455.7 2097152.0 145277.1 154956.0 132223.2 25472.0 17648.7 31 3.881 4 0.879 4.760

524288.0 524288.0 128274.5 0.0 1048576.0 188297.3 2097152.0 153637.1 156492.0 133124.1 25984.0 17845.1 32 4.094 4 0.879 4.973

524288.0 524288.0 0.0 131936.0 1048576.0 209391.3 2097152.0 162141.1 158412.0 134064.2 26624.0 18036.6 33 4.280 4 0.879 5.159

524288.0 524288.0 0.0 131936.0 1048576.0 753862.6 2097152.0 162141.1 158412.0 134064.2 26624.0 18036.6 33 4.280 4 0.879 5.159

524288.0 524288.0 0.0 131936.0 1048576.0 1048576.0 2097152.0 162141.1 158412.0 134064.2 26624.0 18036.6 34 4.280 4 0.879 5.159

524288.0 524288.0 130706.5 0.0 1048576.0 998073.1 2097152.0 169997.1 160076.0 134939.6 27008.0 18208.7 34 4.471 4 0.879 5.350

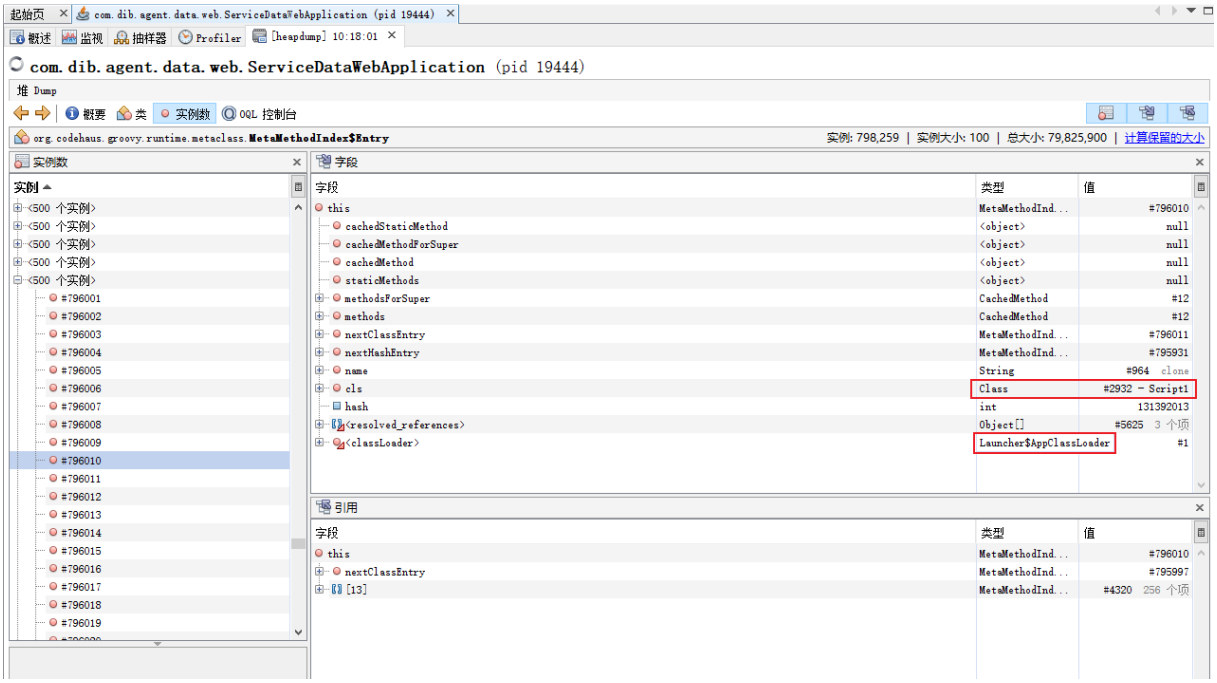
```
c1@Bank-AI-Credit-5142:/opt/app$ jstat -gc 1 3000 100
 S0C    S1C    S0U    S1U      EC       EU       OC         OU       MC     MU    CCSC   CCSU   YGC     YGCT    FGC    FGCT     GCT
524288.0 524288.0 56732.2  0.0   1048576.0 728456.8 2097152.0   70303.2   129484.0 118255.2 17408.0 14625.1     14    0.832   4      0.879    1.711
524288.0 524288.0  0.0   66044.2 1048576.0 973298.0 2097152.0   70311.2   131276.0 119199.0 17920.0 14833.9     15    0.939   4      0.879    1.818
524288.0 524288.0  0.0   84145.3 1048576.0 162792.5 2097152.0   70311.2   134092.0 120840.7 18944.0 15187.7     17    1.211   4      0.879    2.089
524288.0 524288.0  0.0   84145.3 1048576.0 955353.1 2097152.0   70311.2   134092.0 120840.7 18944.0 15187.7     17    1.211   4      0.879    2.089
524288.0 524288.0  0.0   100387.8 1048576.0 20882.8  2097152.0   70359.2   137292.0 122542.2 19840.0 15547.6     19    1.525   4      0.879    2.404
524288.0 524288.0 111071.0  0.0   1048576.0 207947.7 2097152.0   70367.2   139212.0 123496.8 20480.0 15761.3     20    1.699   4      0.879    2.578
524288.0 524288.0  0.0   118995.8 1048576.0 563103.6 2097152.0   70375.2   140364.0 124237.6 20864.0 15911.9     21    1.879   4      0.879    2.758
524288.0 524288.0 128124.5  0.0   1048576.0 752192.3 2097152.0   70375.2   141900.0 125030.9 21376.0 16091.6     22    2.111   4      0.879    2.990
524288.0 524288.0  0.0   133955.8 1048576.0 1019626.6 2097152.0   70415.2   142924.0 125681.0 21632.0 16217.8     23    2.323   4      0.879    3.202
524288.0 524288.0  0.0   131972.7 1048576.0 146340.9 2097152.0   90746.4   145996.0 127400.4 22656.0 16601.6     25    2.751   4      0.879    3.630
524288.0 524288.0 130788.7  0.0   1048576.0 460386.4 2097152.0   100258.4  147532.0 128185.2 23168.0 16764.8     26    2.944   4      0.879    3.823
524288.0 524288.0  0.0   132605.6 1048576.0 774897.9 2097152.0   109602.4  148940.0 128964.8 23552.0 16933.1     27    3.141   4      0.879    4.019
524288.0 524288.0 127794.2 75002.0 1048576.0 1048576.0 2097152.0   125010.4  150476.0 129742.1 24064.0 17104.9     29    3.324   4      0.879    4.203
524288.0 524288.0 130151.5  0.0   1048576.0 398304.3 2097152.0   135413.1  153548.0 131441.2 25088.0 17471.7     30    3.695   4      0.879    4.574
524288.0 524288.0  0.0   129402.0 1048576.0 356455.7 2097152.0   145277.1  154956.0 132223.2 25472.0 17648.7     31    3.881   4      0.879    4.760
524288.0 524288.0 128274.5  0.0   1048576.0 188297.3 2097152.0   153637.1  156492.0 133124.1 25984.0 17845.1     32    4.094   4      0.879    4.973
524288.0 524288.0  0.0   131936.0 1048576.0 209391.3 2097152.0   162141.1  158412.0 134064.2 26624.0 18036.6     33    4.280   4      0.879    5.159
524288.0 524288.0  0.0   131936.0 1048576.0 753862.6 2097152.0   162141.1  158412.0 134064.2 26624.0 18036.6     33    4.280   4      0.879    5.159
524288.0 524288.0  0.0   131936.0 1048576.0 1048576.0 2097152.0   162141.1  158412.0 134064.2 26624.0 18036.6     34    4.280   4      0.879    5.159
524288.0 524288.0 130706.5  0.0   1048576.0 998073.1 2097152.0   169997.1  160076.0 134939.6 27008.0 18208.7     34    4.471   4      0.879    5.350
524288.0 524288.0  0.0   123456.0 1048576.0 792557.9 2097152.0   180701.1  161228.0 135728.3 27392.0 18367.1     35    4.643   4      0.879    5.522
524288.0 524288.0 126304.0  0.0   1048576.0   81.9   2097152.0   188693.1  162764.0 136577.1 27904.0 18525.7     36    4.832   4      0.879    5.711
524288.0 524288.0 126304.0  0.0   1048576.0 396402.0 2097152.0   188693.1  162764.0 136577.1 27904.0 18525.7     36    4.832   4      0.879    5.711
524288.0 524288.0  0.0   118802.5 1048576.0   0.0    2097152.0   197877.1  163968.0 137242.2 28288.0 18659.6     37    5.006   5      0.879    5.885
524288.0 524288.0  0.0    0.0   1048576.0 248836.6 2097152.0   312862.8  163968.0 137175.6 28288.0 18649.3     37    5.006   5      3.846    8.851
524288.0 524288.0 7651.1  0.0   1048576.0 395507.6 2097152.0   312870.8  164992.0 137750.0 28544.0 18771.4     38    5.027   5      3.846    8.873
524288.0 524288.0 17571.1  0.0   1048576.0 41742.0  2097152.0   312878.8  166732.0 138694.7 29056.0 18968.1     40    5.081   5      3.846    8.927
524288.0 524288.0  0.0   22467.1 1048576.0 647017.0 2097152.0   312886.8  167244.0 139104.3 29312.0 19050.4     41    5.118   5      3.846    8.963
524288.0 524288.0  0.0   32251.9 1048576.0 230156.5 2097152.0   312886.8  168652.0 139910.7 29696.0 19214.0     43    5.209   5      3.846    9.055
524288.0 524288.0 37115.9  0.0   1048576.0 900434.6 2097152.0   312886.8  169420.0 140355.5 29952.0 19294.0     44    5.265   5      3.846    9.111
```

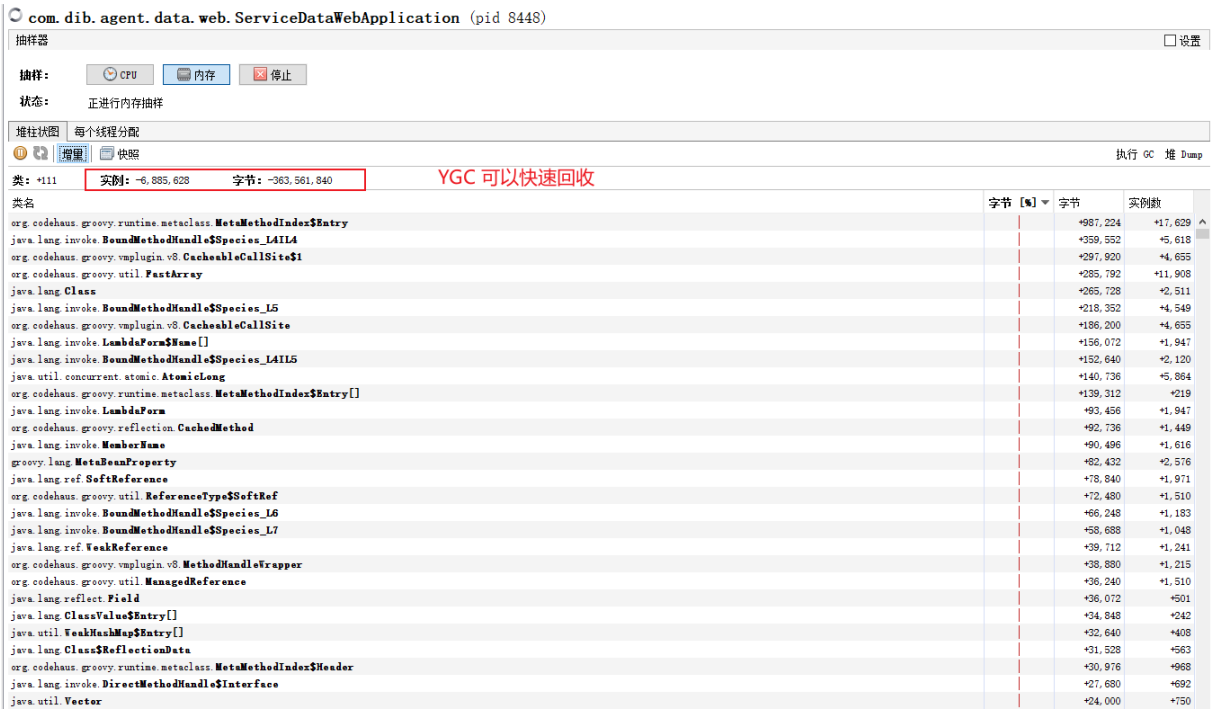
上面是GC日志。

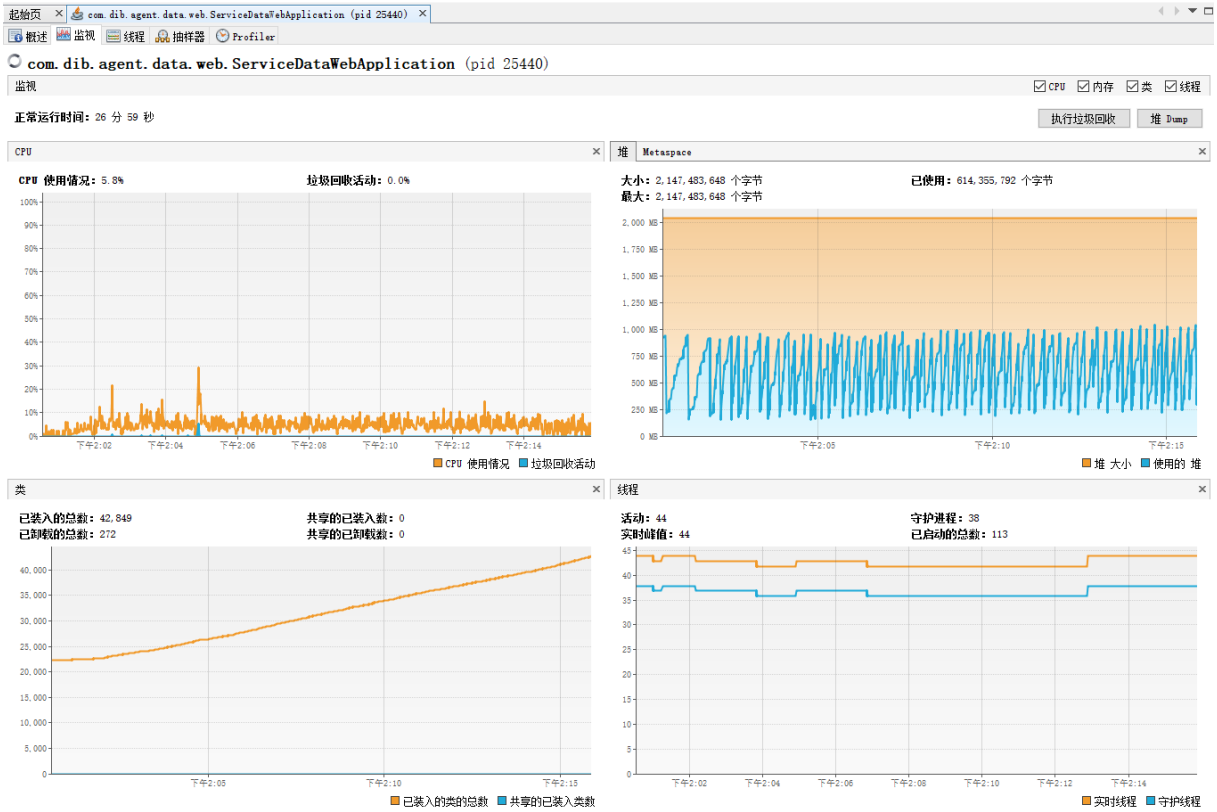
有没有可能，是因为初期新生代放不下的对象，直接晋升到老年代
后面一直 YGC ，因为老年代用到了新生代的对象，导致 YGC 无法清理干净
导致 E 区的占用迅速飙升
E 区不够用时，又有新对象进入了老年代
由此恶性循环，导致老年代也很快被塞满，导致频繁 Full GC

EU增加速度过快，导致YGC非常频繁，每次YGC，OU都在增长

```
c1@Bank-AI-Credit-5142:/opt/app$ jstat -gc 1 200 20
 S0C    S1C    S0U    S1U      EC       EU        OC         OU       MC     MU    CCSC   CCSU   YGC     YGCT    FGC    FGCT     GCT
524288.0 524288.0  0.0   67878.4 1048576.0 167008.7 2097152.0  1379351.9  328012.0 235243.2 80768.0 38053.9    291   30.534   6      9.715   40.249
524288.0 524288.0  0.0   67878.4 1048576.0 250513.0 2097152.0  1379351.9  328012.0 235243.2 80768.0 38053.9    291   30.534   6      9.715   40.249
524288.0 524288.0  0.0   67878.4 1048576.0 354896.7 2097152.0  1379351.9  328012.0 235243.2 80768.0 38053.9    291   30.534   6      9.715   40.249
524288.0 524288.0  0.0   67878.4 1048576.0 459278.4 2097152.0  1379351.9  328012.0 235243.2 80768.0 38053.9    291   30.534   6      9.715   40.249
524288.0 524288.0  0.0   67878.4 1048576.0 563660.1 2097152.0  1379351.9  328012.0 235243.2 80768.0 38053.9    291   30.534   6      9.715   40.249
524288.0 524288.0  0.0   67878.4 1048576.0 668041.9 2097152.0  1379351.9  328012.0 235243.2 80768.0 38053.9    291   30.534   6      9.715   40.249
524288.0 524288.0  0.0   67878.4 1048576.0 751552.3 2097152.0  1379351.9  328012.0 235243.2 80768.0 38053.9    291   30.534   6      9.715   40.249
524288.0 524288.0  0.0   67878.4 1048576.0 855933.9 2097152.0  1379351.9  328012.0 235243.2 80768.0 38053.9    291   30.534   6      9.715   40.249
524288.0 524288.0  0.0   67878.4 1048576.0 939440.5 2097152.0  1379351.9  328012.0 235243.2 80768.0 38053.9    291   30.534   6      9.715   40.249
524288.0 524288.0  0.0   67878.4 1048576.0 1043822.2 2097152.0  1379351.9  328012.0 235243.2 80768.0 38053.9    291   30.534   6      9.715   40.249
524288.0 524288.0 67878.4  0.0   1048576.0 41826.2  2097152.0  1383799.9  328396.0 235595.0 80896.0 38124.4    292   30.630   6      9.715   40.345
524288.0 524288.0 67878.4  0.0   1048576.0 146376.5 2097152.0  1383799.9  328396.0 235595.0 80896.0 38124.4    292   30.630   6      9.715   40.345
524288.0 524288.0 67878.4  0.0   1048576.0 250927.1 2097152.0  1383799.9  328396.0 235595.0 80896.0 38124.4    292   30.630   6      9.715   40.345
524288.0 524288.0 67878.4  0.0   1048576.0 355484.1 2097152.0  1383799.9  328396.0 235595.0 80896.0 38124.4    292   30.630   6      9.715   40.345
524288.0 524288.0 67878.4  0.0   1048576.0 460034.7 2097152.0  1383799.9  328396.0 235595.0 80896.0 38124.4    292   30.630   6      9.715   40.345
524288.0 524288.0 67878.4  0.0   1048576.0 543676.5 2097152.0  1383799.9  328396.0 235595.0 80896.0 38124.4    292   30.630   6      9.715   40.345
524288.0 524288.0 67878.4  0.0   1048576.0 648225.0 2097152.0  1383799.9  328396.0 235595.0 80896.0 38124.4    292   30.630   6      9.715   40.345
524288.0 524288.0 67878.4  0.0   1048576.0 773686.1 2097152.0  1383799.9  328396.0 235595.0 80896.0 38124.4    292   30.630   6      9.715   40.345
524288.0 524288.0 67878.4  0.0   1048576.0 773692.2 2097152.0  1383799.9  328396.0 235595.0 80896.0 38124.4    292   30.630   6      9.715   40.345
524288.0 524288.0 67878.4  0.0   1048576.0 857351.6 2097152.0  1383799.9  328396.0 235595.0 80896.0 38124.4    292   30.630   6      9.715   40.345
c1@Bank-AI-Credit-5142:/opt/app$ jstat -gc 1 200 20
 S0C    S1C    S0U    S1U      EC       EU        OC         OU       MC     MU    CCSC   CCSU   YGC     YGCT    FGC    FGCT     GCT
524288.0 524288.0  0.0   65467.2 1048576.0 627295.6 2097152.0  1546800.0  351308.0 249216.5 88192.0 40805.3    329   34.212   6      9.715   43.928
524288.0 524288.0  0.0   65467.2 1048576.0 710933.4 2097152.0  1546800.0  351308.0 249216.5 88192.0 40805.3    329   34.212   6      9.715   43.928
524288.0 524288.0  0.0   65467.2 1048576.0 815481.9 2097152.0  1546800.0  351308.0 249216.5 88192.0 40805.3    329   34.212   6      9.715   43.928
524288.0 524288.0  0.0   65467.2 1048576.0 899121.8 2097152.0  1546800.0  351308.0 249216.5 88192.0 40805.3    329   34.212   6      9.715   43.928
524288.0 524288.0  0.0   65467.2 1048576.0 1003669.8 2097152.0  1546800.0  351308.0 249216.5 88192.0 40805.3    329   34.212   6      9.715   43.928
524288.0 524288.0 66395.2  0.0   1048576.0   0.0    2097152.0  1550336.0  352076.0 249669.0 88448.0 40900.8    330   34.308   6      9.715   44.023
524288.0 524288.0 66395.2  0.0   1048576.0 104665.0 2097152.0  1550336.0  352076.0 249669.0 88448.0 40900.8    330   34.308   6      9.715   44.023
524288.0 524288.0 66395.2  0.0   1048576.0 188393.5 2097152.0  1550336.0  352076.0 249669.0 88448.0 40900.8    330   34.308   6      9.715   44.023
524288.0 524288.0 66395.2  0.0   1048576.0 293054.6 2097152.0  1550336.0  352076.0 249669.0 88448.0 40900.8    330   34.308   6      9.715   44.023
524288.0 524288.0 66395.2  0.0   1048576.0 397713.9 2097152.0  1550336.0  352076.0 249669.0 88448.0 40900.8    330   34.308   6      9.715   44.023
524288.0 524288.0 66395.2  0.0   1048576.0 481442.1 2097152.0  1550336.0  352076.0 249669.0 88448.0 40900.8    330   34.308   6      9.715   44.023
524288.0 524288.0 66395.2  0.0   1048576.0 586101.3 2097152.0  1550336.0  352076.0 249669.0 88448.0 40900.8    330   34.308   6      9.715   44.023
524288.0 524288.0 66395.2  0.0   1048576.0 690760.4 2097152.0  1550336.0  352076.0 249669.0 88448.0 40900.8    330   34.308   6      9.715   44.023
524288.0 524288.0 66395.2  0.0   1048576.0 774489.1 2097152.0  1550336.0  352076.0 249669.0 88448.0 40900.8    330   34.308   6      9.715   44.023
524288.0 524288.0 66395.2  0.0   1048576.0 879148.2 2097152.0  1550336.0  352076.0 249669.0 88448.0 40900.8    330   34.308   6      9.715   44.023
524288.0 524288.0 66395.2  0.0   1048576.0 983806.9 2097152.0  1550336.0  352076.0 249669.0 88448.0 40900.8    330   34.308   6      9.715   44.023
```




com.dib.agent.data.web.ServiceDataWebApplication (pid 8448)
抽样器
设置
抽样：
CPU
内存
停止
状态：
正进行内存抽样
堆柱状图
每个线程分配
增量
快照
执行 GC
堆 Dump
类：+111
实例：-6,885,628
字节：-363,561,840
YGC 可以快速回收
类名
字节 [%]
字节
实例数
org.codehaus.groovy.runtime.metaclass.MetaMethodIndex$Entry +987,224 +17,629
java.lang.invoke.BoundMethodHandle$Species_LAIL4 +359,552 +5,618
org.codehaus.groovy.vmplugin.v8.CacheableCallSite$1 +297,920 +4,655
org.codehaus.groovy.util.FastArray +285,792 +11,908
java.lang.Class +265,728 +2,511
java.lang.invoke.BoundMethodHandle$Species_L5 +218,352 +4,549
org.codehaus.groovy.vmplugin.v8.CacheableCallSite +186,200 +4,655
java.lang.invoke.LambdaForm$Name[] +156,072 +1,947
java.lang.invoke.BoundMethodHandle$Species_LAIL5 +152,640 +2,120
java.util.concurrent.atomic.AtomicLong +140,736 +5,864
org.codehaus.groovy.runtime.metaclass.MetaMethodIndex$Entry[] +139,312 +219
java.lang.invoke.LambdaForm +93,456 +1,947
org.codehaus.groovy.reflection.CachedMethod +92,736 +1,449
java.lang.invoke.MemberName +90,496 +1,616
groovy.lang.MetaBeanProperty +82,432 +2,576
java.lang.ref.SoftReference +78,840 +1,971
org.codehaus.groovy.util.ReferenceType$SoftRef +72,480 +1,510
java.lang.invoke.BoundMethodHandle$Species_L6 +66,248 +1,183
java.lang.invoke.BoundMethodHandle$Species_L7 +58,688 +1,048
java.lang.ref.WeakReference +39,712 +1,241
org.codehaus.groovy.vmplugin.v8.MethodHandleWrapper +38,880 +1,215
org.codehaus.groovy.util.ManagedReference +36,240 +1,510
java.lang.reflect.Field +36,072 +501
java.lang.ClassValue$Entry[] +34,848 +242
java.util.WeakHashMap$Entry[] +32,640 +408
java.lang.Class$ReflectionData +31,528 +563
org.codehaus.groovy.runtime.metaclass.MetaMethodIndex$Header +30,976 +968
java.lang.invoke.DirectMethodHandle$Interface +27,680 +692
java.util.Vector +24,000 +750


起始页
com.dib.agent.data.web.ServiceDataWebApplication (pid 25440)
概述
监视
线程
抽样器
Profiler
com.dib.agent.data.web.ServiceDataWebApplication (pid 25440)
监视
CPU
内存
类
线程
正常运行时间：26 分 59 秒
执行垃圾回收
堆 Dump
CPU
CPU 使用情况：5.8%
垃圾回收活动：0.0%
CPU 使用情况
垃圾回收活动
堆
Metaspace
大小：2,147,483,648 个字节
最大：2,147,483,648 个字节
已使用：614,355,792 个字节
堆 大小
使用的 堆
类
已装入的总数：42,849
已卸载的总数：272
共享的已装入数：0
共享的已卸载数：0
已装入的类的总数
共享的已装入类数
线程
活动：44
实时峰值：44
守护进程：38
已启动的总数：113
实时线程
守护线程

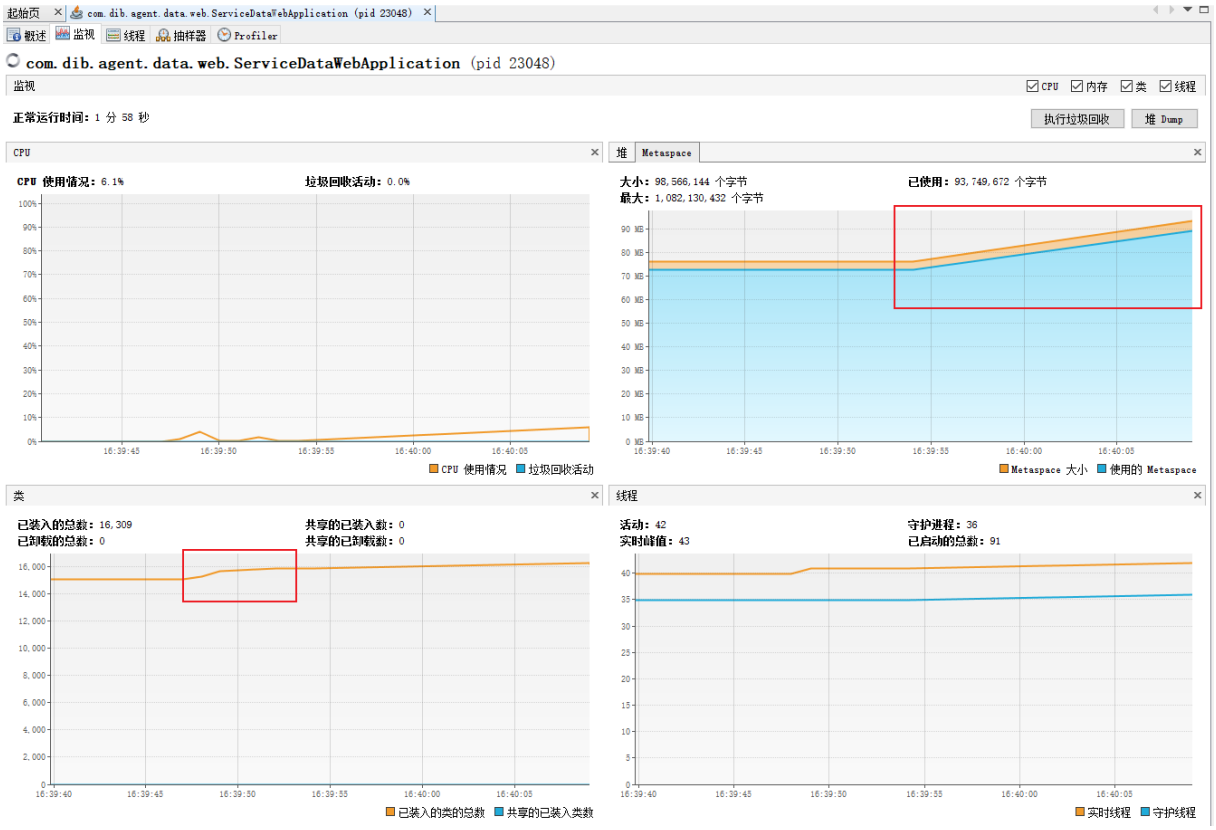
full gc 全部都是由 metaspace 到达阈值触发的

剩余2个要解决的问题：

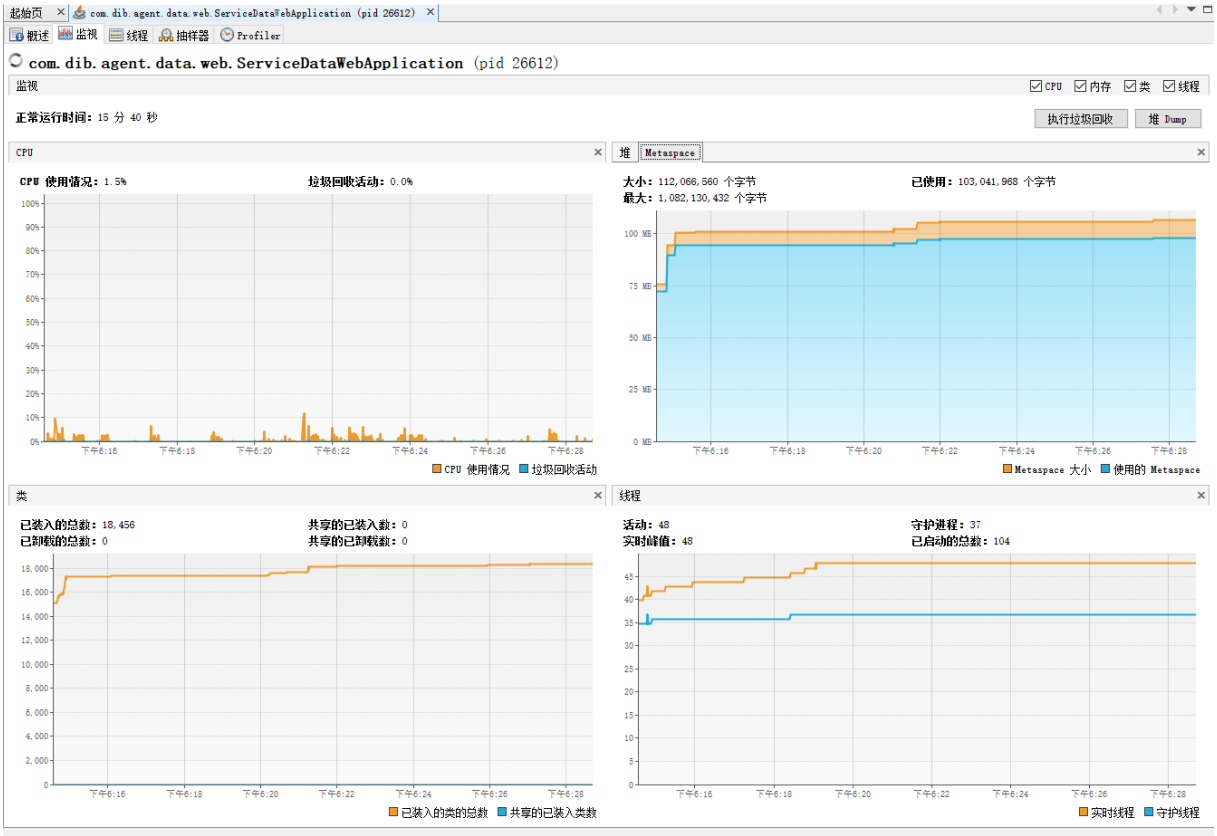
1、设置 meataspace 相关 jvm 参数

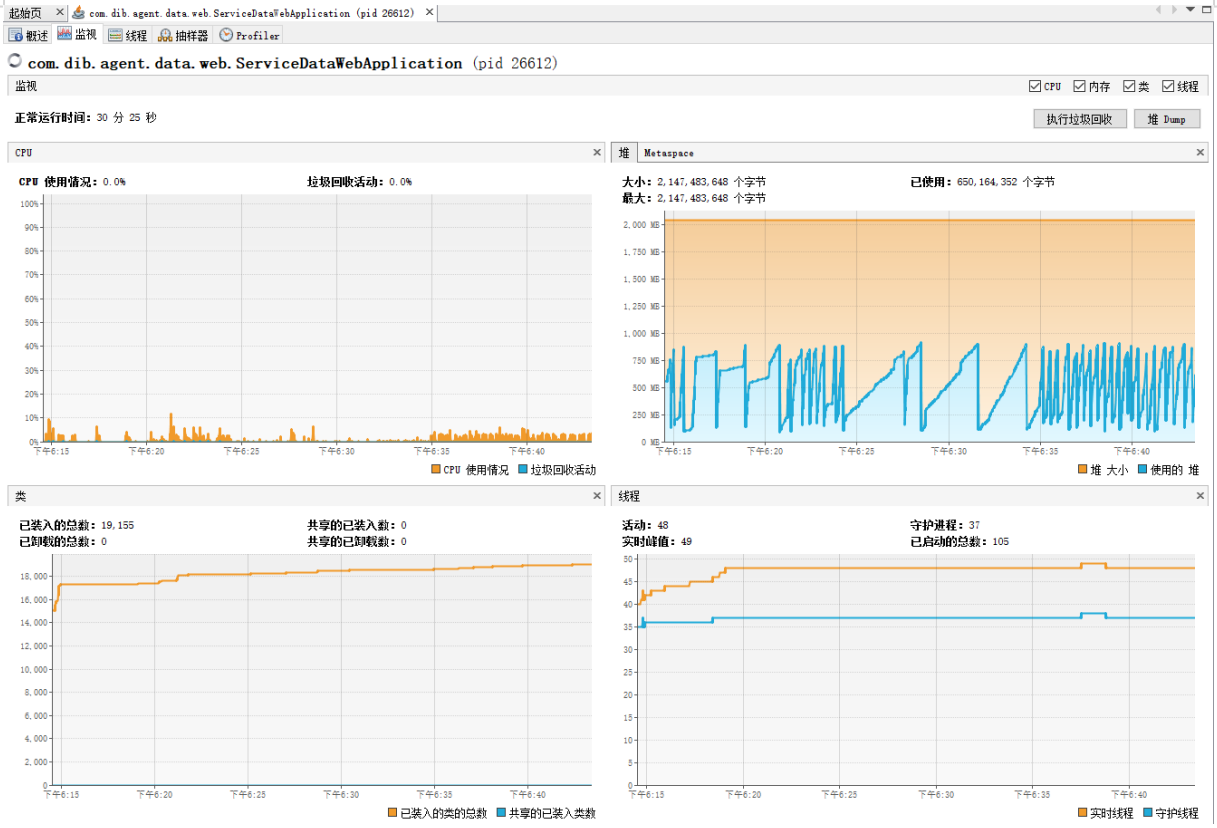
2、在脚本中 groovyShell 主动释放 classLoader，避免 Script.class 无法卸载

INDEX_19531496



```
PS E:\Project_DIB> jstat -gcutil  26612  3000 100
  S0     S1     E      O      M     CCS    YGC     YGCT    FGC    FGCT    CGC    CGCT     GCT
 54.31   0.00  54.58   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  54.71   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  54.82   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  54.96   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  57.02   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  57.13   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  59.18   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  59.32   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  59.45   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  61.49   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  61.63   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  61.76   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  63.82   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  63.93   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  65.99   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  66.12   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  66.25   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  68.29   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  70.34   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  70.49   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  80.23   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  82.26   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
 54.31   0.00  82.40   3.71  92.08  86.91     18    0.631     0    0.000      -       -    0.631
```




起始页
com.dib.agent.data.web.ServiceDataWebApplication (pid 26612)
概述
监视
线程
抽样器
Profiler
com.dib.agent.data.web.ServiceDataWebApplication (pid 26612)
监视
CPU
内存
类
线程
正常运行时间：30 分 25 秒
执行垃圾回收
堆 Dump
CPU
CPU 使用情况：0.0%
垃圾回收活动：0.0%
CPU 使用情况
垃圾回收活动
堆
Metaspace
大小：2,147,483,648 个字节
最大：2,147,483,648 个字节
已使用：650,164,352 个字节
堆 大小
使用的 堆
类
已装入的总数：19,155
已卸载的总数：0
共享的已装入数：0
共享的已卸载数：0
已装入的类的总数
共享的已装入类数
线程
活动：48
实时峰值：49
守护进程：37
已启动的总数：105
实时线程
守护线程

起始页
com.dib.agent.data.web.ServiceDataWebApplication (pid 26612)
概述
监视
线程
抽样器
Profiler
com.dib.agent.data.web.ServiceDataWebApplication (pid 26612)
监视
CPU
内存
类
线程
正常运行时间：51 分 14 秒
执行垃圾回收
堆 Dump
CPU
CPU 使用情况：0.0%
垃圾回收活动：0.0%
CPU 使用情况
垃圾回收活动
堆
Metaspace
大小：2,147,483,648 个字节
最大：2,147,483,648 个字节
已使用：821,668,488 个字节
堆 大小
使用的 堆
类
已装入的总数：19,449
已卸载的总数：0
共享的已装入数：0
共享的已卸载数：0
已装入的类的总数
共享的已装入类数
线程
活动：47
实时峰值：49
守护进程：37
已启动的总数：106
实时线程
守护线程

```
PS E:\Project_DIB> jstat -gcutil  26612  3000 100
  S0     S1     E      O      M     CCS    YGC     YGCT    FGC    FGCT     CGC    CGCT       GCT
  3.85   0.00  93.36  10.48  89.46  84.02     66    0.993     0    0.000     -       -     0.993
  0.00   3.65   8.15  10.48  89.28  84.05     67    0.998     0    0.000     -       -     0.998
  0.00   3.65  24.27  10.48  89.28  84.05     67    0.998     0    0.000     -       -     0.998
  0.00   3.65  40.43  10.48  89.28  84.05     67    0.998     0    0.000     -       -     0.998
  0.00   3.65  56.54  10.48  89.28  84.05     67    0.998     0    0.000     -       -     0.998
  0.00   3.65  60.69  10.48  89.28  84.05     67    0.998     0    0.000     -       -     0.998
  0.00   3.65  60.80  10.48  89.28  84.05     67    0.998     0    0.000     -       -     0.998
  0.00   3.65  62.96  10.48  89.28  84.05     67    0.998     0    0.000     -       -     0.998
  0.00   3.65  65.06  10.48  89.28  84.05     67    0.998     0    0.000     -       -     0.998
  0.00   3.65  87.17  10.48  89.28  84.05     67    0.998     0    0.000     -       -     0.998
  3.04   0.00   9.97  10.49  89.25  83.52     68    1.003     0    0.000     -       -     1.003
  3.04   0.00  10.14  10.49  89.25  83.52     68    1.003     0    0.000     -       -     1.003
  3.04   0.00  26.13  10.49  89.25  83.52     68    1.003     0    0.000     -       -     1.003
  3.04   0.00  40.16  10.49  89.25  83.52     68    1.003     0    0.000     -       -     1.003
  3.04   0.00  46.25  10.49  89.25  83.52     68    1.003     0    0.000     -       -     1.003
  3.04   0.00  46.34  10.49  89.25  83.52     68    1.003     0    0.000     -       -     1.003
  3.04   0.00  46.47  10.49  89.25  83.52     68    1.003     0    0.000     -       -     1.003
  3.04   0.00  48.59  10.49  89.25  83.52     68    1.003     0    0.000     -       -     1.003
  3.04   0.00  48.72  10.49  89.25  83.52     68    1.003     0    0.000     -       -     1.003
  3.04   0.00  48.85  10.49  89.25  83.52     68    1.003     0    0.000     -       -     1.003
  3.04   0.00  48.98  10.49  89.25  83.52     68    1.003     0    0.000     -       -     1.003
  3.04   0.00  51.10  10.49  89.25  83.52     68    1.003     0    0.000     -       -     1.003
  3.04   0.00  51.25  10.49  89.25  83.52     68    1.003     0    0.000     -       -     1.003
  3.04   0.00  51.38  10.49  89.25  83.52     68    1.003     0    0.000     -       -     1.003
  3.04   0.00  53.50  10.49  89.25  83.52     68    1.003     0    0.000     -       -     1.003
  3.04   0.00  53.63  10.49  89.25  83.52     68    1.003     0    0.000     -       -     1.003
```

```
sipf@pingjia-VMware-Virtual-web-2:~$ docker exec -it dib-agent-service-data-web /bin/bash
c1@pingjia-VMware-Virtual-web-2:/opt/app$ jstat -gc 1 1000 10
 S0C    S1C    S0U    S1U      EC       EU        OC         OU       MC     MU    CCSC   CCSU   YGC     YGCT    FGC    FGCT     GCT
10752.0 11264.0  0.0   3216.4 675840.0 592207.4 2796544.0  2772026.4  526380.0 333829.5 156204.0 56308.6   3942   96.410 1114  2321.526 2417.937
10752.0 11264.0  0.0   3216.4 675840.0 592207.4 2796544.0  2772026.4  526380.0 333829.5 156204.0 56308.6   3942   96.410 1114  2321.526 2417.937
10752.0 11264.0  0.0   3216.4 675840.0 592211.9 2796544.0  2772026.4  526380.0 333829.5 156204.0 56308.6   3942   96.410 1114  2321.526 2417.937
10752.0 11264.0  0.0   3216.4 675840.0 592211.9 2796544.0  2772026.4  526380.0 333829.5 156204.0 56308.6   3942   96.410 1114  2321.526 2417.937
10752.0 11264.0  0.0   3216.4 675840.0 592214.0 2796544.0  2772026.4  526380.0 333829.5 156204.0 56308.6   3942   96.410 1114  2321.526 2417.937
10752.0 11264.0  0.0   3216.4 675840.0 592214.0 2796544.0  2772026.4  526380.0 333829.5 156204.0 56308.6   3942   96.410 1114  2321.526 2417.937
10752.0 11264.0  0.0   3216.4 675840.0 637956.4 2796544.0  2772026.4  526380.0 333829.5 156204.0 56308.6   3942   96.410 1114  2321.526 2417.937
10752.0 11264.0  0.0   3216.4 675840.0 668530.7 2796544.0  2772026.4  526380.0 333829.5 156204.0 56308.6   3942   96.410 1114  2321.526 2417.937
10752.0 11264.0  0.0   3216.4 675840.0 668530.7 2796544.0  2772026.4  526380.0 333829.5 156204.0 56308.6   3942   96.410 1114  2321.526 2417.937
10752.0 10240.0 3648.4  0.0   677888.0  6905.2  2796544.0  2772058.4  526380.0 333832.1 156204.0 56308.6   3943   96.435 1114  2321.526 2417.962
c1@pingjia-VMware-Virtual-web-2:/opt/app$
```

```
c1@Bank-AI-Credit-5142:/opt/app$ jstat -gcutil 1 1000 10
  S0     S1     E      O      M     CCS    YGC     YGCT    FGC    FGCT     GCT
  0.00 100.00  44.60  72.21  89.03  83.93   1969   23.491     0    0.000   23.491
  0.00 100.00  52.26  72.21  89.03  83.93   1969   23.491     0    0.000   23.491
  0.00 100.00  58.36  72.21  89.03  83.93   1969   23.491     0    0.000   23.491
  0.00 100.00  65.57  72.21  89.03  83.93   1969   23.491     0    0.000   23.491
  0.00 100.00  71.68  72.21  89.03  83.93   1969   23.491     0    0.000   23.491
  0.00 100.00  83.66  72.21  89.03  83.93   1969   23.491     0    0.000   23.491
  0.00 100.00  89.29  72.21  89.03  83.93   1969   23.491     0    0.000   23.491
  0.00 100.00  96.07  72.21  89.03  83.93   1969   23.491     0    0.000   23.491
100.00   0.00   3.54  72.21  89.03  83.93   1970   23.653     0    0.000   23.653
100.00   0.00  20.48  72.21  89.03  83.93   1970   23.653     0    0.000   23.653
c1@Bank-AI-Credit-5142:/opt/app$ jstat -gcutil 1 1000 100
  S0     S1     E      O      M     CCS    YGC     YGCT    FGC    FGCT     GCT
  2.36   0.00  10.68  12.48  86.64  79.72   2002   26.303     1    0.473   26.776
  2.36   0.00  54.99  12.48  86.64  79.72   2002   26.303     1    0.473   26.776
  0.00   3.24   2.00  12.48  86.66  79.77   2003   26.313     1    0.473   26.786
  0.00   3.24  40.13  12.48  86.66  79.77   2003   26.313     1    0.473   26.786
  0.00   3.24  95.48  12.48  86.66  79.77   2003   26.313     1    0.473   26.786
  3.97   0.00  24.58  12.49  86.66  79.77   2004   26.323     1    0.473   26.796
  3.97   0.00  82.15  12.49  86.66  79.77   2004   26.323     1    0.473   26.796
  0.00   3.19  21.44  12.49  86.66  79.77   2005   26.333     1    0.473   26.806
  0.00   3.19  74.92  12.49  86.66  79.77   2005   26.333     1    0.473   26.806
  4.52   0.00  12.94  12.50  86.66  79.77   2006   26.344     1    0.473   26.817
  4.52   0.00  76.38  12.50  86.66  79.77   2006   26.344     1    0.473   26.817
  0.00   2.41   8.66  12.51  86.66  79.77   2007   26.353     1    0.473   26.826
  0.00   2.41  61.35  12.51  86.66  79.77   2007   26.353     1    0.473   26.826
  0.00   2.41  95.13  12.51  86.66  79.77   2007   26.353     1    0.473   26.826
```

> 🤔 那么这两种方式在性能上还有区别吗？它们的差异到底体现在哪里？

✅ 回答：是的，两者在这一步的行为确实是一样的 —— 都创建了 Script 实例并注入 Binding。

但它们**前面的编译过程完全不同**，而这个编译过程才是性能和资源管理的关键所在！

🔍 一、两者的本质区别在于"如何获取 `Class<? extends Script>`"

步骤	方法1（`GroovyShell.parse`）	方法2（`GroovyClassLoader.parseClass`）
获取 Class 的方式	通过已有的 Script 实例获取类	直接从 Groovy 源码字符串编译成类
是否创建 Script 实例?	✅ 是（多余步骤）	❌ 否
是否绑定上下文?	✅ 是（可能带 Binding、ClassLoader 状态）	❌ 否
编译效率	⚠️ 较低（多做了实例化）	✅ 更高（轻量级编译）
是否适合缓存?	❌ 不推荐（状态不干净）	✅ 推荐（线程安全）

✅ 二、举个生活化的例子帮助理解

想象你要做 5000 个蛋糕：

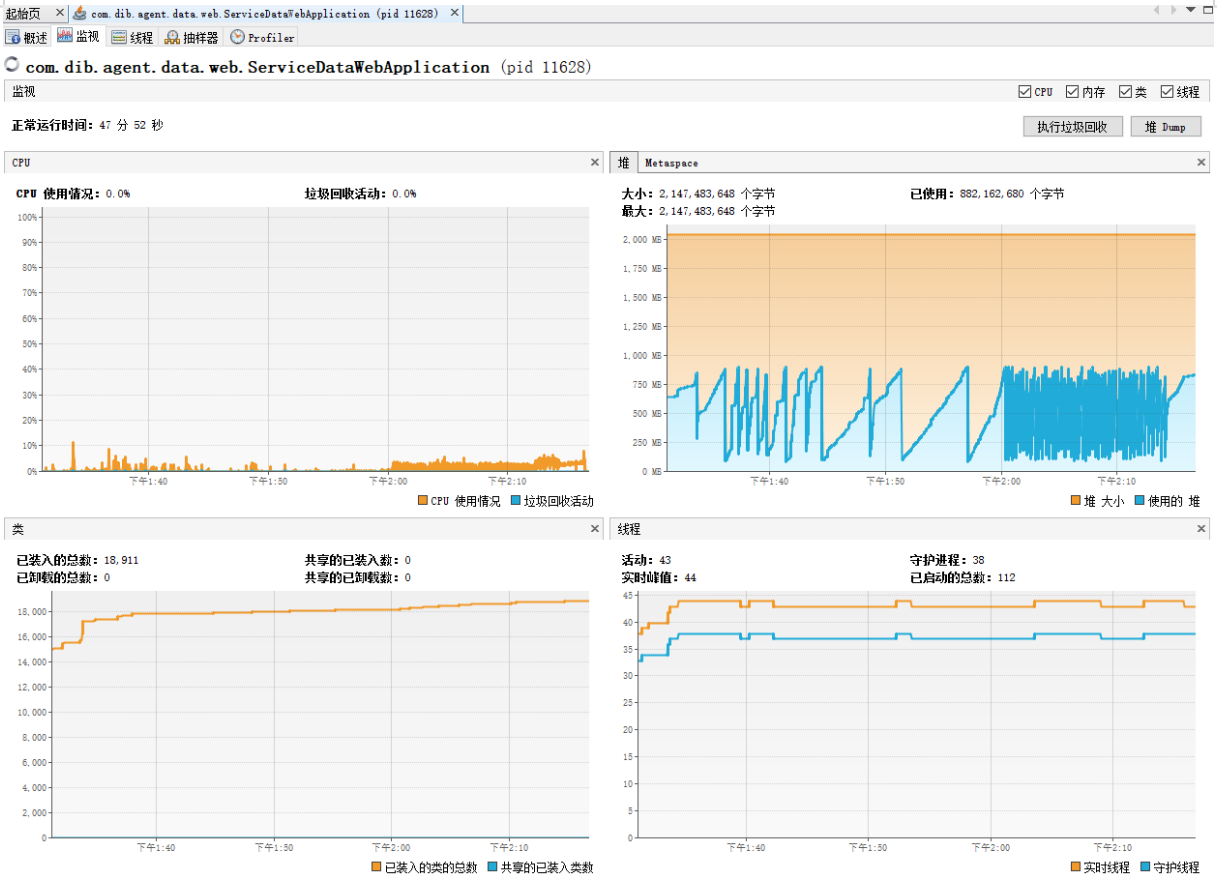
- 方法1：每次做蛋糕都要先烤一个"示例蛋糕"（Script 实例），再用它去复制模具；
- 方法2：直接根据配方（脚本文本）制作模具（Class），不需要先烤一个蛋糕。

虽然最后都要用模具做蛋糕（创建 Script 实例），但 **前者的准备工作更慢、更重、更浪费资源。**

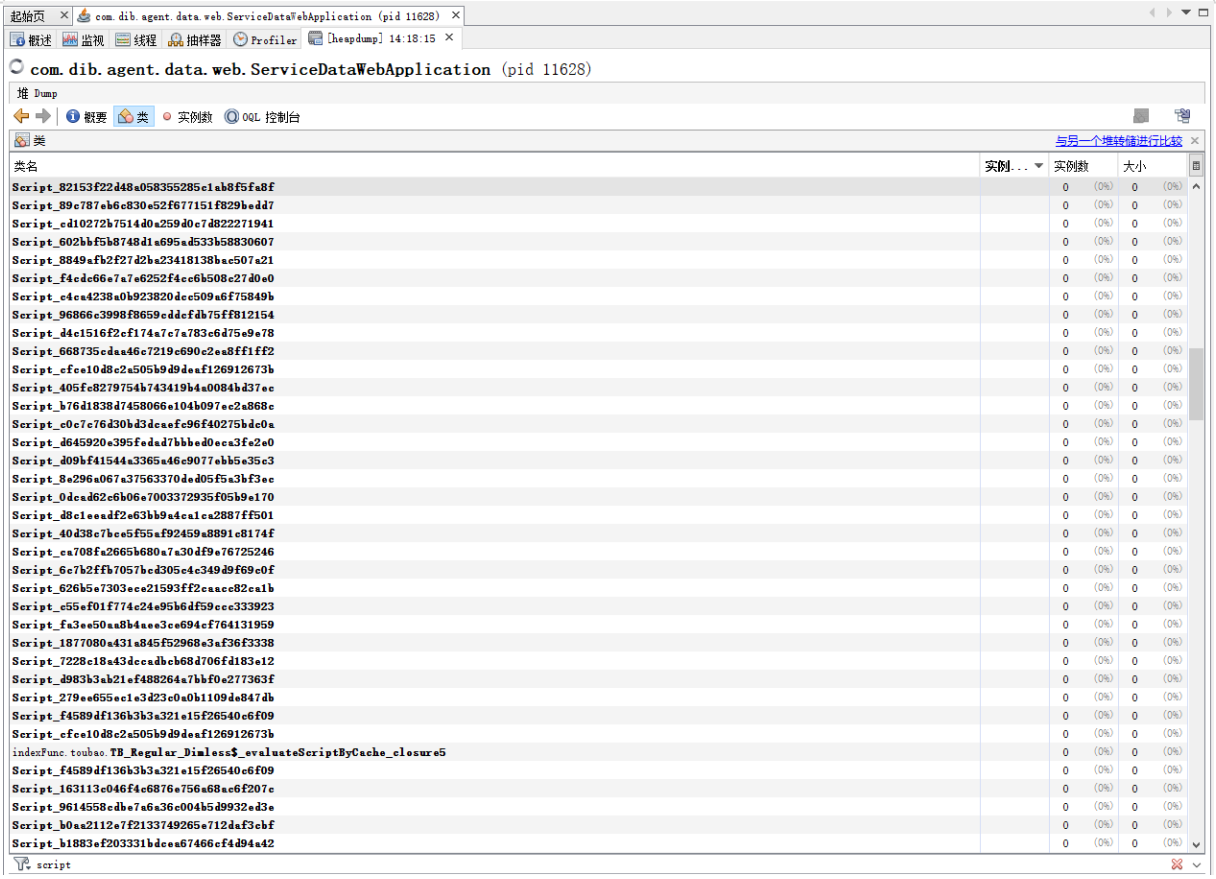
✅ 三、所以真正的性能差异在于哪?

💡 **差异主要体现在：**

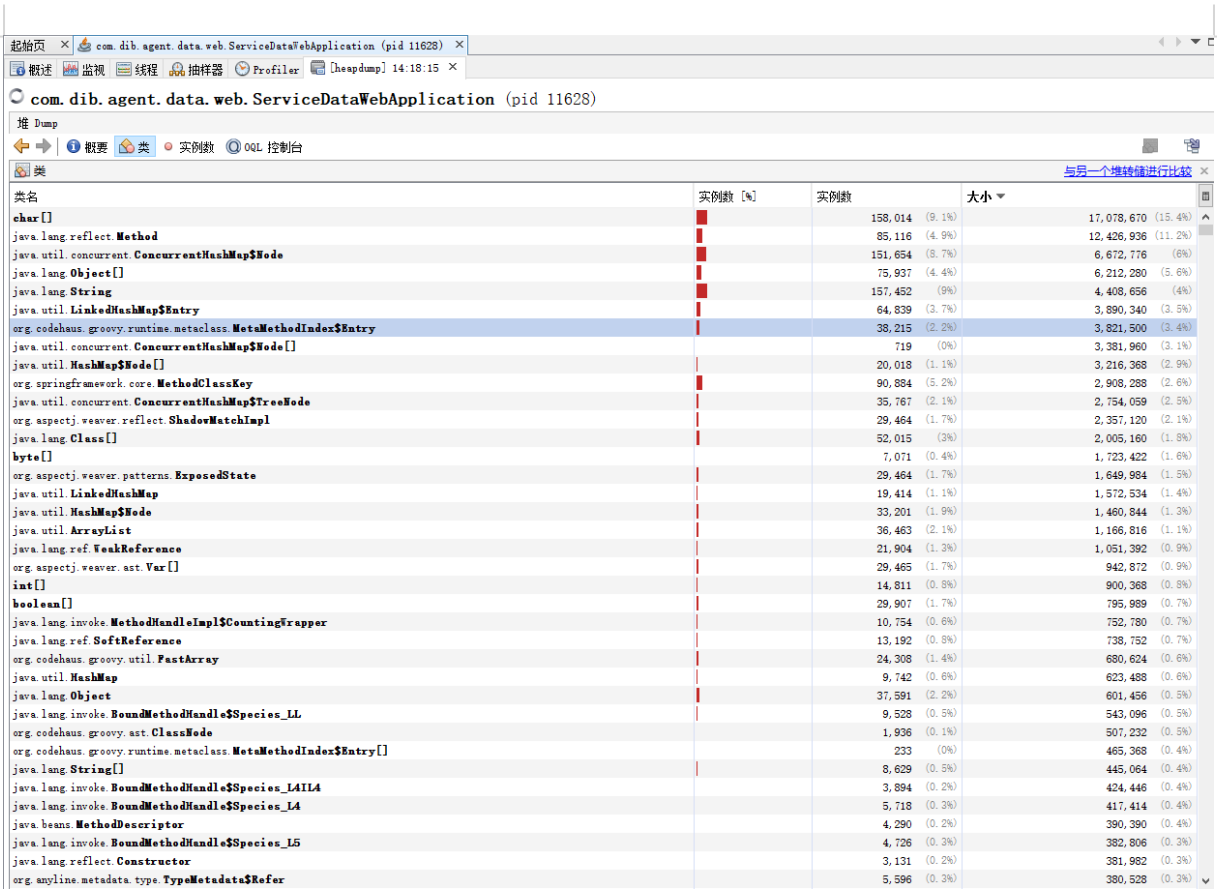
项目	方法1	方法2
每次处理脚本时是否需要创建 Script 实例?	✅ 是（额外开销）	❌ 否
是否绑定了不必要的上下文?	✅ 是（GroovyShell 带 Binding / ClassLoader）	❌ 否
是否容易造成内存泄漏?	✅ 是（共享 GroovyShell 时）	❌ 否（临时 ClassLoader）
是否适合缓存 Script 类?	❌ 否（状态不清）	✅ 是（纯净类对象）
总体性能	⚠️ 较差	✅ 更好




```
PS E:\Project_DIB> jstat -gcutil  11628  3000 100
  S0     S1     E      O      M     CCS    YGC     YGCT    FGC    FGCT    CGC    CGCT     GCT
  3.84   0.00  52.45   9.15  90.27  85.95     56    0.876     0    0.000     -       -    0.876
  3.84   0.00  70.56   9.15  90.27  85.95     56    0.876     0    0.000     -       -    0.876
  3.84   0.00  72.68   9.15  90.27  85.95     56    0.876     0    0.000     -       -    0.876
  3.84   0.00  90.79   9.15  90.27  85.95     56    0.876     0    0.000     -       -    0.876
  0.00   3.95  20.10   9.16  90.27  85.95     57    0.881     0    0.000     -       -    0.881
  0.00   3.95  36.22   9.16  90.27  85.95     57    0.881     0    0.000     -       -    0.881
  0.00   3.95  54.34   9.16  90.27  85.95     57    0.881     0    0.000     -       -    0.881
  0.00   3.95  58.46   9.16  90.27  85.95     57    0.881     0    0.000     -       -    0.881
  0.00   3.95  74.57   9.16  90.27  85.95     57    0.881     0    0.000     -       -    0.881
  0.00   3.95  90.85   9.16  90.27  85.95     57    0.881     0    0.000     -       -    0.881
  3.78   0.00  22.12   9.17  90.28  85.96     58    0.886     0    0.000     -       -    0.886
  3.78   0.00  38.24   9.17  90.28  85.96     58    0.886     0    0.000     -       -    0.886
  3.78   0.00  42.37   9.17  90.28  85.96     58    0.886     0    0.000     -       -    0.886
  3.78   0.00  46.48   9.17  90.28  85.96     58    0.886     0    0.000     -       -    0.886
  3.78   0.00  62.70   9.17  90.28  85.96     58    0.886     0    0.000     -       -    0.886
  3.78   0.00  82.79   9.17  90.28  85.96     58    0.886     0    0.000     -       -    0.886
  0.00   3.64   0.00   9.18  90.29  85.97     59    0.891     0    0.000     -       -    0.891
  0.00   3.64  22.18   9.18  90.29  85.97     59    0.891     0    0.000     -       -    0.891
  0.00   3.64  28.27   9.18  90.29  85.97     59    0.891     0    0.000     -       -    0.891
  0.00   3.64  30.37   9.18  90.29  85.97     59    0.891     0    0.000     -       -    0.891
```


起始页
com.dib.agent.data.web.ServiceDataWebApplication (pid 11628)
概述
监视
线程
抽样器
Profiler
[heapdump] 14:18:15
com.dib.agent.data.web.ServiceDataWebApplication (pid 11628)
堆 Dump
概要
类
实例数
OQL 控制台
类
与另一个堆转储进行比较
类名
实例...
实例数
大小
Script_82153f22d48a058355285e1ab8f5fa8f
Script_89c787eb6c830e52f677151f829bedd7
Script_cd10272b7514d0a259d0c7d822271941
Script_602bbf5b8748d1a695ad533b58830607
Script_8849afb2f27d2ba23418138bac507a21
Script_f4cdc66e7a7e6252f4cc6b508c27d0e0
Script_c4ca4238a0b923820dcc509a6f75849b
Script_96866c3998f8659cddcfdb75ff812154
Script_d4c1516f2cf174a7c7a783c6d75e9e78
Script_668735cdaa46c7219c690c2ea8ff1ff2
Script_cfce10d8c2a505b9d9deaf126912673b
Script_405fc8279754b743419b4a0084bd37ec
Script_b76d1838d7458066e104b097ec2a868c
Script_c0c7c76d30bd3dcaefc96f40275bdc0a
Script_d645920e395fedad7bbbed0eca3fe2e0
Script_d09bf41544a3365a46c9077ebb5e35c3
Script_8e296a067a37563370ded05f5a3bf3ec
Script_0dcad62c6b06e7003372935f05b9e170
Script_d8c1eeadf2e63bb9a4ca1ca2887ff501
Script_40d38c7bce5f55af92459a8891c8174f
Script_ca708fa2665b680a7a30df9e76725246
Script_6c7b2ffb7057bcd305c4c349d9f69c0f
Script_626b5a7303ece21593ff2caacc82ca1b
Script_c55ef01f774c24e95b6df59ccc333923
Script_fa3ee50aa8b4aee3ce694cf764131959
Script_1877080a431a845f52968e3af36f3338
Script_7228c18a43dccadbcb68d706fd183e12
Script_d983b3ab21ef488264a7bbf0e277363f
Script_279ee655ec1e3d23c0a0b1109de847db
Script_f4589df136b3b3a321e15f26540c6f09
Script_cfce10d8c2a505b9d9deaf126912673b
indexFunc.toubao.TB_Regular_Dimless$_evaluateScriptByCache_closure5
Script_f4589df136b3b3a321e15f26540c6f09
Script_163113c046f4c6876e756a68ac6f207c
Script_9614558cdbe7a6a36c004b5d9932ed3e
Script_b0aa2112e7f2133749265e712daf3cbf
Script_b1883ef203331bdcea67466cf4d94a42
script

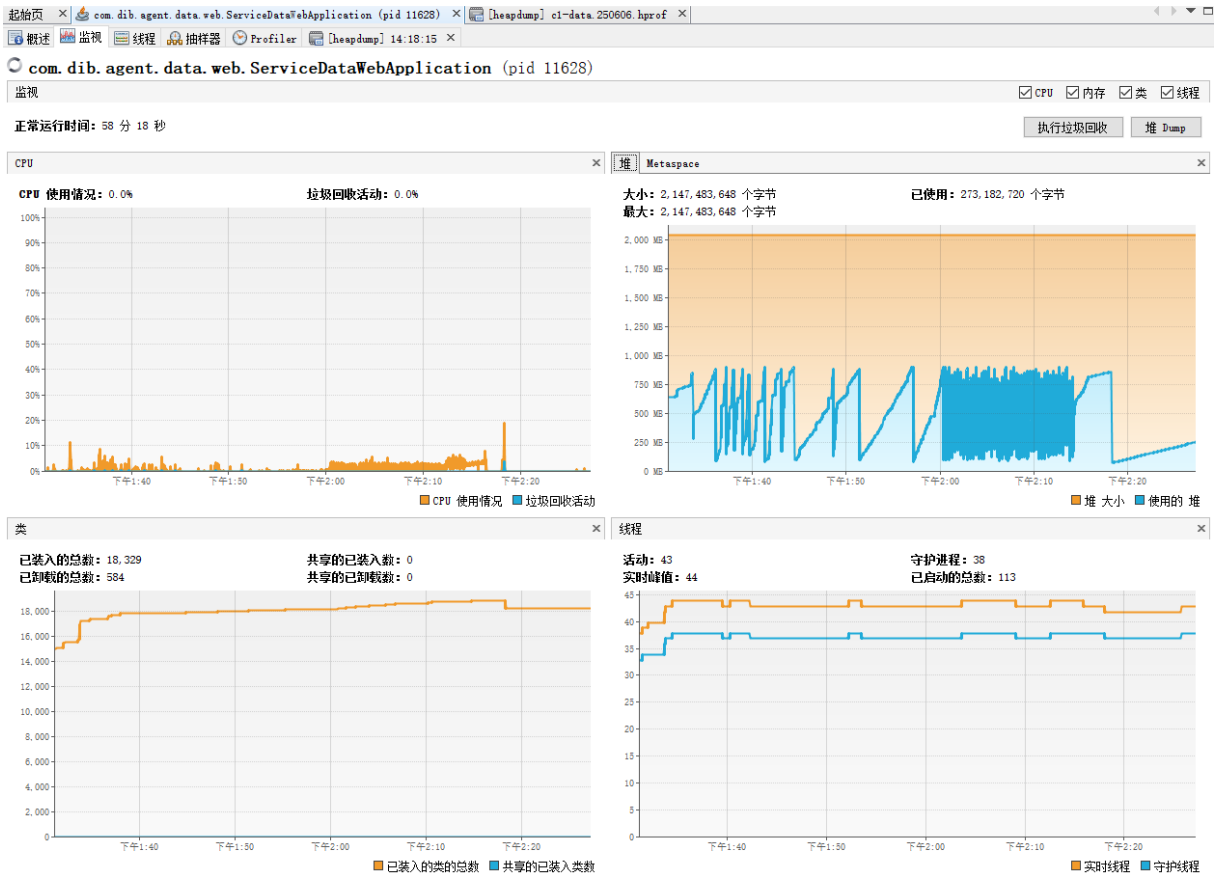
优化后

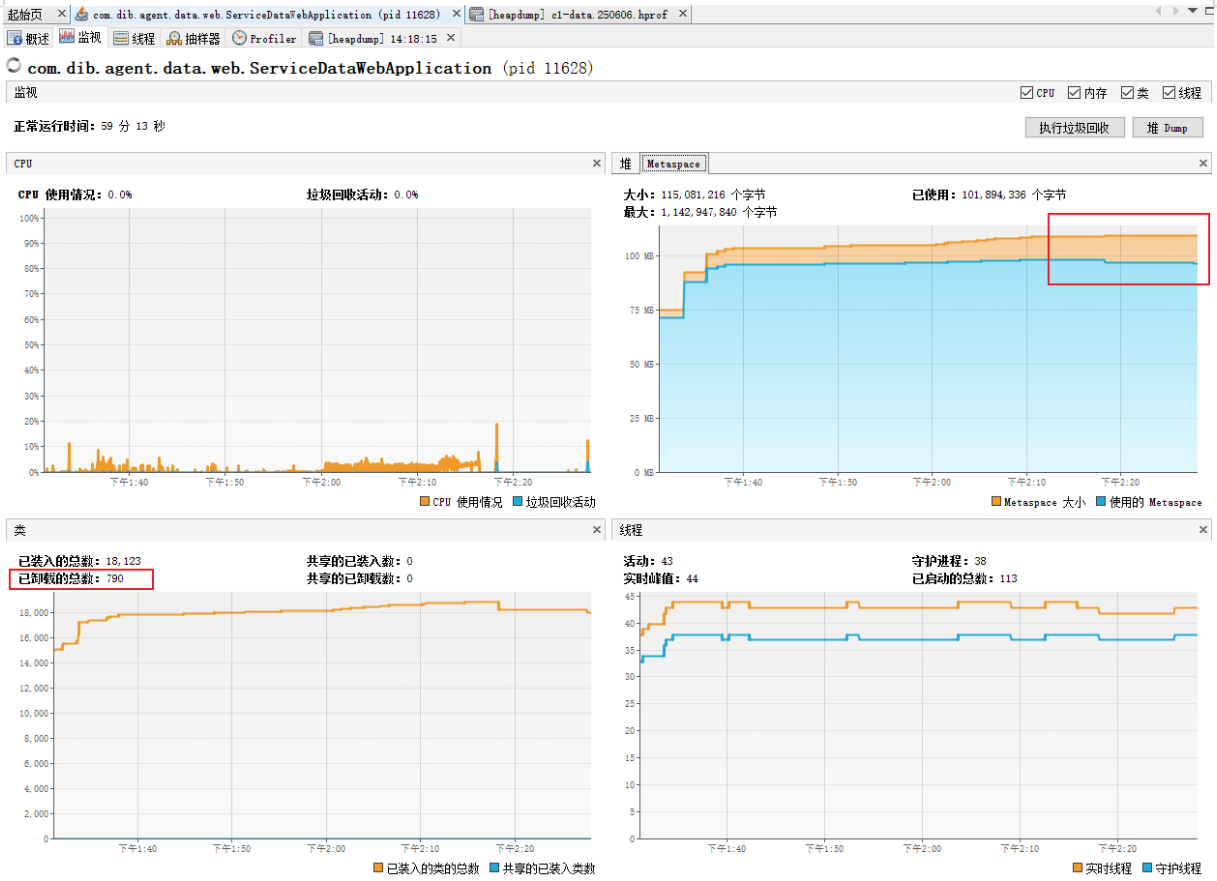


类名	实例数 [%]	实例数		大小	
char[]		158,014	(9.1%)	17,078,670	(15.4%)
java.lang.reflect.Method		85,116	(4.9%)	12,426,936	(11.2%)
java.util.concurrent.ConcurrentHashMap$Node		151,654	(8.7%)	6,672,776	(6%)
java.lang.Object[]		75,937	(4.4%)	6,212,280	(5.6%)
java.lang.String		157,452	(9%)	4,408,656	(4%)
java.util.LinkedHashMap$Entry		64,839	(3.7%)	3,890,340	(3.5%)
org.codehaus.groovy.runtime.metaclass.MetaMethodIndex$Entry		38,215	(2.2%)	3,821,500	(3.4%)
java.util.concurrent.ConcurrentHashMap$Node[]		719	(0%)	3,381,960	(3.1%)
java.util.HashMap$Node[]		20,018	(1.1%)	3,216,368	(2.9%)
org.springframework.core.MethodClassKey		90,884	(5.2%)	2,908,288	(2.6%)
java.util.concurrent.ConcurrentHashMap$TreeNode		35,767	(2.1%)	2,754,059	(2.5%)
org.aspectj.weaver.reflect.ShadowMatchImpl		29,464	(1.7%)	2,357,120	(2.1%)
java.lang.Class[]		52,015	(3%)	2,005,160	(1.8%)
byte[]		7,071	(0.4%)	1,723,422	(1.6%)
org.aspectj.weaver.patterns.ExposedState		29,464	(1.7%)	1,649,984	(1.5%)
java.util.LinkedHashMap		19,414	(1.1%)	1,572,534	(1.4%)
java.util.HashMap$Node		33,201	(1.9%)	1,460,844	(1.3%)
java.util.ArrayList		36,463	(2.1%)	1,166,816	(1.1%)
java.lang.ref.WeakReference		21,904	(1.3%)	1,051,392	(0.9%)
org.aspectj.weaver.ast.Var[]		29,465	(1.7%)	942,872	(0.9%)
int[]		14,811	(0.8%)	900,368	(0.8%)
boolean[]		29,907	(1.7%)	795,989	(0.7%)
java.lang.invoke.MethodHandleImpl$CountingWrapper		10,754	(0.6%)	752,780	(0.7%)
java.lang.ref.SoftReference		13,192	(0.8%)	738,752	(0.7%)
org.codehaus.groovy.util.FastArray		24,308	(1.4%)	680,624	(0.6%)
java.util.HashMap		9,742	(0.6%)	623,488	(0.6%)
java.lang.Object		37,591	(2.2%)	601,456	(0.5%)
java.lang.invoke.BoundMethodHandle$Species_LL		9,528	(0.5%)	543,096	(0.5%)
org.codehaus.groovy.ast.ClassNode		1,936	(0.1%)	507,232	(0.5%)
org.codehaus.groovy.runtime.metaclass.MetaMethodIndex$Entry[]		233	(0%)	465,368	(0.4%)
java.lang.String[]		8,629	(0.5%)	445,064	(0.4%)
java.lang.invoke.BoundMethodHandle$Species_L4IL4		3,894	(0.2%)	424,446	(0.4%)
java.lang.invoke.BoundMethodHandle$Species_L4		5,718	(0.3%)	417,414	(0.4%)
java.beans.MethodDescriptor		4,290	(0.2%)	390,390	(0.4%)
java.lang.invoke.BoundMethodHandle$Species_L5		4,726	(0.3%)	382,806	(0.3%)
java.lang.reflect.Constructor		3,131	(0.2%)	381,982	(0.3%)
org.anyline.metadata.type.TypeMetadata$Refer		5,596	(0.3%)	380,528	(0.3%)

dump触发full gc

```
  0.00   4.09  93.09   9.22  90.35  86.08     63     0.911     0     0.000     -         -     0.911
  0.00   4.09  93.21   9.22  90.35  86.08     63     0.911     0     0.000     -         -     0.911
  0.00   4.09  93.34   9.22  90.35  86.08     63     0.911     0     0.000     -         -     0.911
  0.00   4.09  93.46   9.22  90.35  86.08     63     0.911     0     0.000     -         -     0.911
  0.00   4.09  93.56   9.22  90.35  86.08     63     0.911     0     0.000     -         -     0.911
  0.00   4.09  93.69   9.22  90.35  86.08     63     0.911     0     0.000     -         -     0.911
  0.00   4.09  93.81   9.22  90.35  86.08     63     0.911     0     0.000     -         -     0.911
  0.00   0.00   0.00   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
  0.00   0.00   0.23   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
  0.00   0.00   0.39   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
PS E:\Project_DIB> jstat -gcutil  11628  3000 100
  S0     S1     E      O      M     CCS    YGC     YGCT    FGC    FGCT    CGC    CGCT      GCT
  0.00   0.00   0.47   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
  0.00   0.00   0.61   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
  0.00   0.00   0.70   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
  0.00   0.00   0.85   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
  0.00   0.00   0.94   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
  0.00   0.00   1.09   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
  0.00   0.00   1.17   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
  0.00   0.00   1.32   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
  0.00   0.00   1.40   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
  0.00   0.00   1.56   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
  0.00   0.00   1.64   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
  0.00   0.00   1.79   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
  0.00   0.00   1.87   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
  0.00   0.00   1.96   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
  0.00   0.00   2.13   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
  0.00   0.00   2.21   7.51  88.92  84.15     64     0.915     1     0.327     -         -     1.242
```


起始页
com.dib.agent.data.web.ServiceDataWebApplication (pid 11628)
[heapdump] c1-data.250606.hprof
概述
监视
线程
抽样器
Profiler
[heapdump] 14:18:15
com.dib.agent.data.web.ServiceDataWebApplication (pid 11628)
监视
CPU
内存
类
线程
正常运行时间: 58 分 18 秒
执行垃圾回收
堆 Dump
CPU
CPU 使用情况: 0.0%
垃圾回收活动: 0.0%
CPU 使用情况
垃圾回收活动
堆
Metaspace
大小: 2,147,483,648 个字节
最大: 2,147,483,648 个字节
已使用: 273,182,720 个字节
堆 大小
使用的 堆
类
已装入的总数: 18,329
已卸载的总数: 584
共享的已装入数: 0
共享的已卸载数: 0
已装入的类的总数
共享的已装入类数
线程
活动: 43
实时峰值: 44
守护进程: 38
已启动的总数: 113
实时线程
守护线程


起始页
com.dib.agent.data.web.ServiceDataWebApplication (pid 11628)
[heapdump] c1-data.250606.hprof
概述
监视
线程
抽样器
Profiler
[heapdump] 14:18:15
com.dib.agent.data.web.ServiceDataWebApplication (pid 11628)
监视
CPU
内存
类
线程
正常运行时间：59 分 13 秒
执行垃圾回收
堆 Dump
CPU
CPU 使用情况：0.0%
垃圾回收活动：0.0%
CPU 使用情况
垃圾回收活动
堆
Metaspace
大小：115,081,216 个字节
最大：1,142,947,840 个字节
已使用：101,894,336 个字节
Metaspace 大小
使用的 Metaspace
类
已装入的总数：18,123
已卸载的总数：790
共享的已装入数：0
共享的已卸载数：0
已装入的类的总数
共享的已装入类数
线程
活动：43
实时峰值：44
守护进程：38
已启动的总数：113
实时线程
守护线程

优化前

起始页 | com.dib.agent.data.web.ServiceDataWebApplication (pid 11628) | [heapdump] c1-data.250606.hprof

[heapdump] c1-data.250606.hprof

堆 Dump

概要 | 类 | 实例数 | OQL 控制台

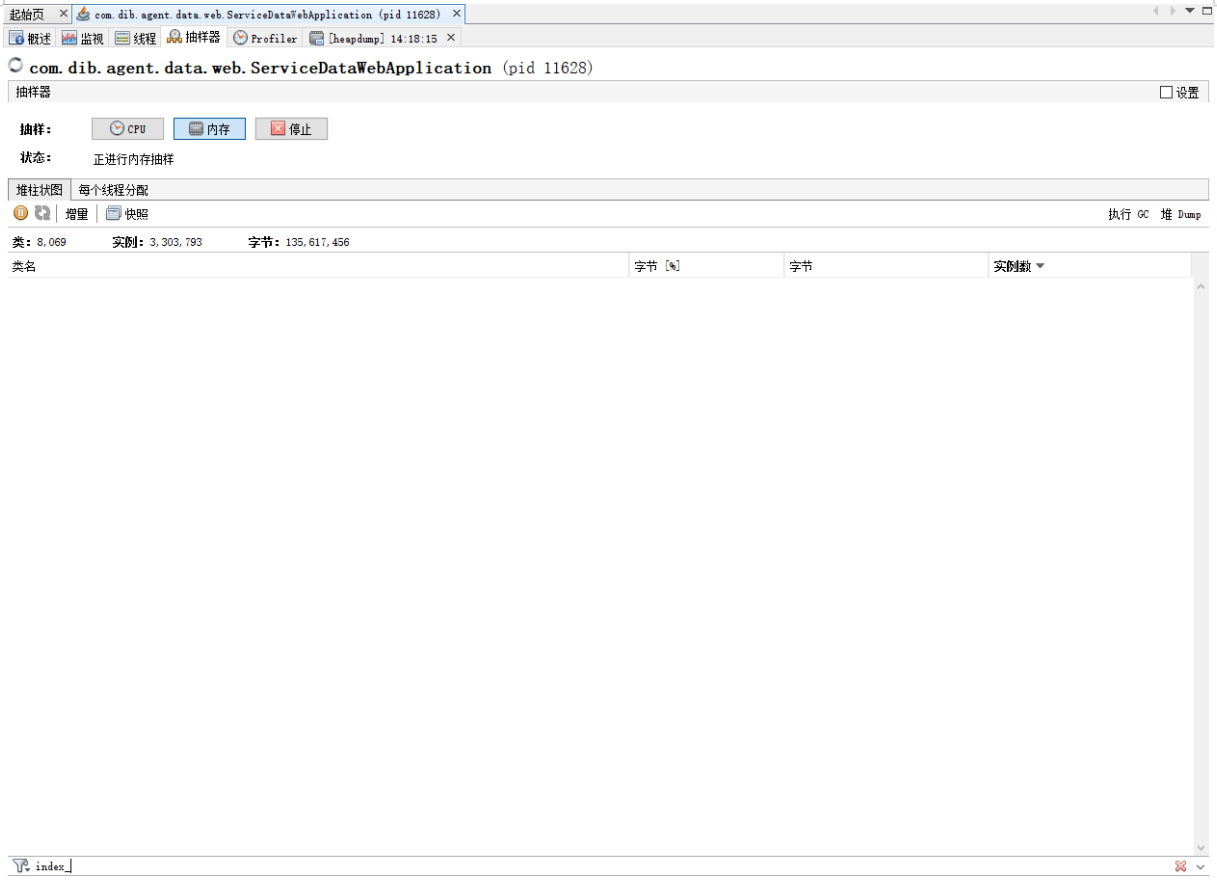
类

与另一个堆转储进行比较

类名	实例数 [%]	实例数	大小
org.codehaus.groovy.runtime.metaclass.MetaMethodIndex$Entry		10,753,647 (16.6%)	1,075,364,700 (24.1%)
java.lang.Object[]		7,989,272 (12.3%)	400,866,216 (9%)
org.codehaus.groovy.util.FastArray		7,204,374 (11.1%)	201,722,472 (4.5%)
java.util.LinkedHashMap$Entry		7,086,520 (10.9%)	425,191,200 (9.5%)
java.lang.invoke.MethodHandleImpl$CountingWrapper		1,714,042 (2.6%)	119,982,940 (2.7%)
java.util.concurrent.locks.AbstractQueuedSynchronizer$Node		1,602,011 (2.5%)	83,304,572 (1.9%)
groovy.lang.MetaBeanProperty		1,600,938 (2.5%)	89,652,528 (2%)
java.lang.Class[]		1,285,356 (2%)	39,898,472 (0.9%)
java.util.HashMap$Node[]		1,221,599 (1.9%)	212,920,600 (4.8%)
java.lang.invoke.BoundMethodHandle$Species_LL		1,219,103 (1.9%)	69,488,871 (1.6%)
java.lang.reflect.Method		1,093,232 (1.7%)	159,611,872 (3.6%)
java.lang.invoke.BoundMethodHandle$Species_L4		1,085,343 (1.7%)	79,230,039 (1.8%)
java.util.LinkedHashMap		996,726 (1.5%)	80,734,806 (1.8%)
java.lang.ref.WeakReference		883,951 (1.4%)	42,429,648 (1%)
int[]		877,811 (1.4%)	85,165,768 (1.9%)
java.util.concurrent.atomic.AtomicLong		735,965 (1.1%)	17,663,160 (0.4%)
java.lang.ref.SoftReference		726,774 (1.1%)	40,699,344 (0.9%)
java.lang.invoke.BoundMethodHandle$Species_L5		682,361 (1.1%)	55,271,241 (1.2%)
java.lang.invoke.BoundMethodHandle$Species_L4IL4		635,600 (1%)	69,280,400 (1.6%)
java.util.ArrayList		605,404 (0.9%)	19,372,928 (0.4%)
char[]		566,319 (0.9%)	77,961,414 (1.7%)
java.lang.String		524,979 (0.8%)	14,699,412 (0.3%)
java.util.HashMap		484,813 (0.7%)	31,028,032 (0.7%)
java.util.concurrent.ConcurrentHashMap$Node		452,349 (0.7%)	19,903,356 (0.4%)
org.codehaus.groovy.util.ReferenceType$SoftRef		413,821 (0.6%)	26,484,544 (0.6%)
org.codehaus.groovy.util.ManagedReference		413,821 (0.6%)	13,242,272 (0.3%)
java.lang.reflect.Constructor		412,034 (0.6%)	50,268,148 (1.1%)
org.codehaus.groovy.vmplugin.v8.CacheableCallSite$1		389,820 (0.6%)	34,693,980 (0.8%)
org.codehaus.groovy.vmplugin.v8.CacheableCallSite		389,820 (0.6%)	24,948,480 (0.6%)
org.codehaus.groovy.vmplugin.v8.MethodHandleWrapper		344,838 (0.5%)	14,138,358 (0.3%)
java.beans.MethodRef		331,868 (0.5%)	13,274,720 (0.3%)
java.util.LinkedHashMap$LinkedEntrySet		287,313 (0.4%)	6,895,512 (0.2%)
java.util.Vector		237,234 (0.4%)	8,540,424 (0.2%)
org.codehaus.groovy.reflection.CachedMethod		235,759 (0.4%)	24,518,936 (0.5%)
java.lang.reflect.Field		234,675 (0.4%)	26,518,275 (0.6%)
java.lang.invoke.BoundMethodHandle$Species_L		233,445 (0.4%)	11,438,805 (0.3%)
org.codehaus.groovy.runtime.metaclass.MetaMethodIndex$Header		230,210 (0.4%)	10,129,240 (0.2%)
java.util.HashMap$Node		226,515 (0.3%)	9,966,660 (0.2%)

full gc 之后， script_ 以及 index_ 动态加载的 clz 都已被卸载

起始页
com.dib.agent.data.web.ServiceDataWebApplication (pid 11628)
概述
监视
线程
抽样器
Profiler
[heapdump] 14:18:15
com.dib.agent.data.web.ServiceDataWebApplication (pid 11628)
监视
CPU
内存
类
线程
正常运行时间：1 小时 06 分 14 秒
执行垃圾回收
堆 Dump
CPU
CPU 使用情况：0.0%
垃圾回收活动：0.0%
CPU 使用情况
垃圾回收活动
堆
Metaspace
大小：2,147,483,648 个字节
最大：2,147,483,648 个字节
已使用：100,721,024 个字节
堆 大小
使用的 堆
类
已装入的总数：18,055
已卸载的总数：858
共享的已装入数：0
共享的已卸载数：0
已装入的类的总数
共享的已装入类数
线程
活动：42
实时峰值：44
守护进程：37
已启动的总数：113
实时线程
守护线程



再次执行指标，可以看到 clz=index_

起始页 | com.dib.agent.data.web.ServiceDataWebApplication (pid 11628)

概述 | 监视 | 线程 | 抽样器 | Profiler | [heapdump] 14:18:15

com.dib.agent.data.web.ServiceDataWebApplication (pid 11628)

抽样器 | 设置

抽样： CPU | 内存 | 停止

状态： 正进行内存抽样

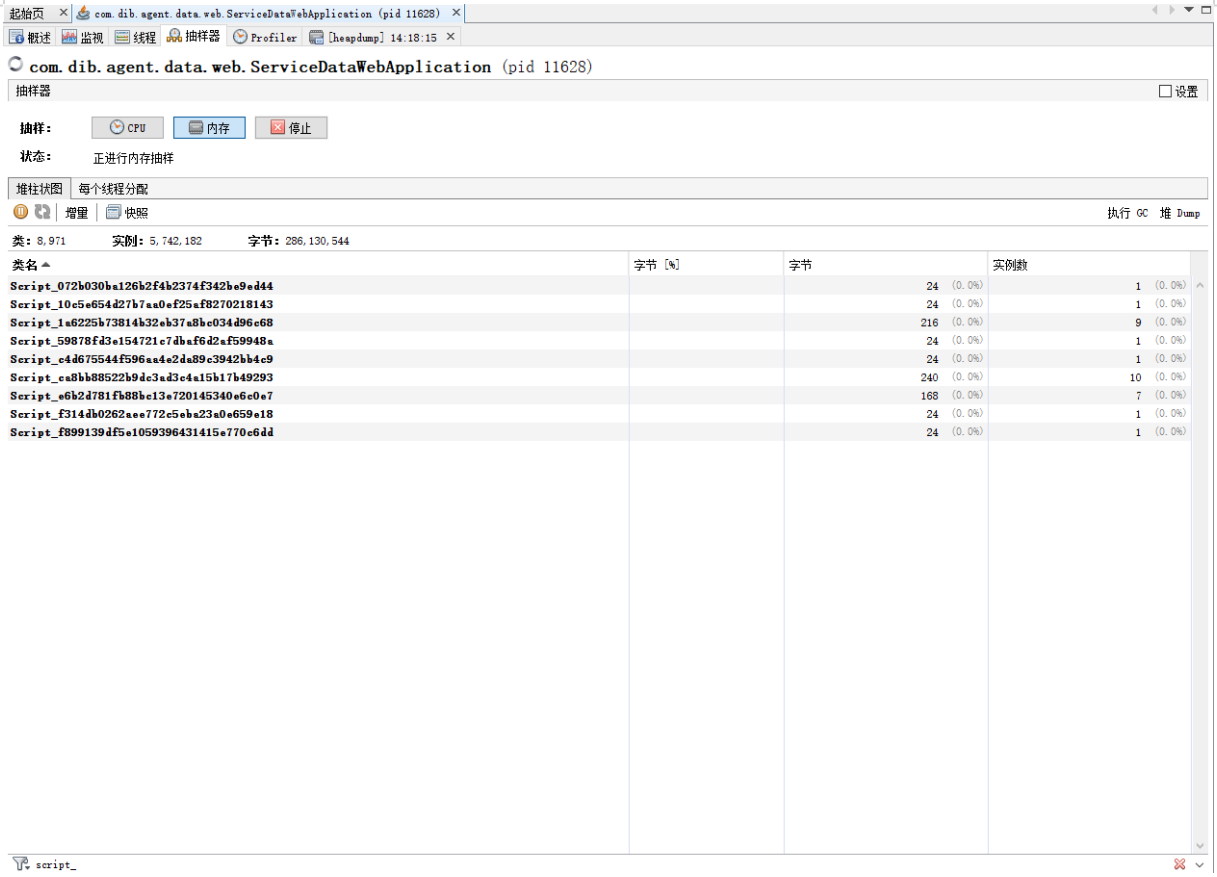
堆柱状图 | 每个线程分配

增量 | 快照 | 执行 GC | 堆 Dump

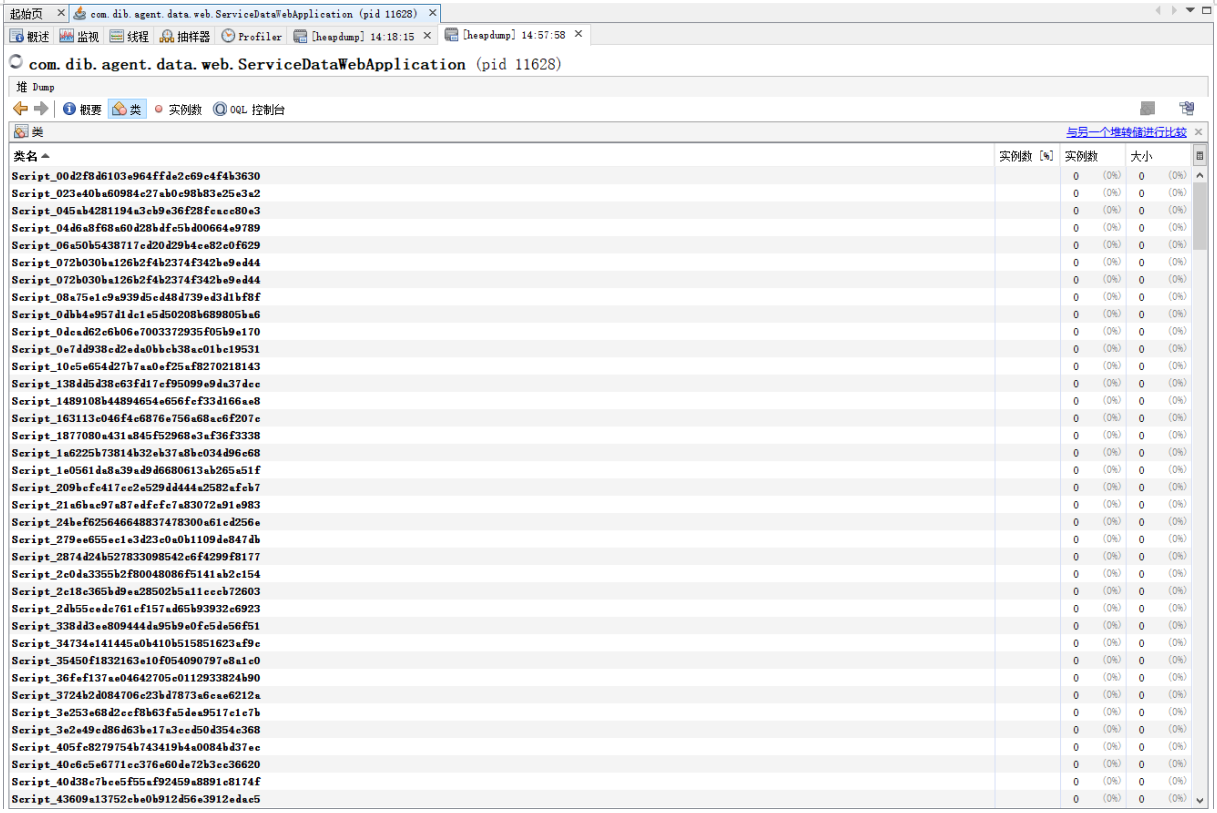
类：8,950 实例：4,483,776 字节：231,370,096

类名	字节 [%]	字节	实例数
INDEX_02262565		24 (0.0%)	1 (0.0%)

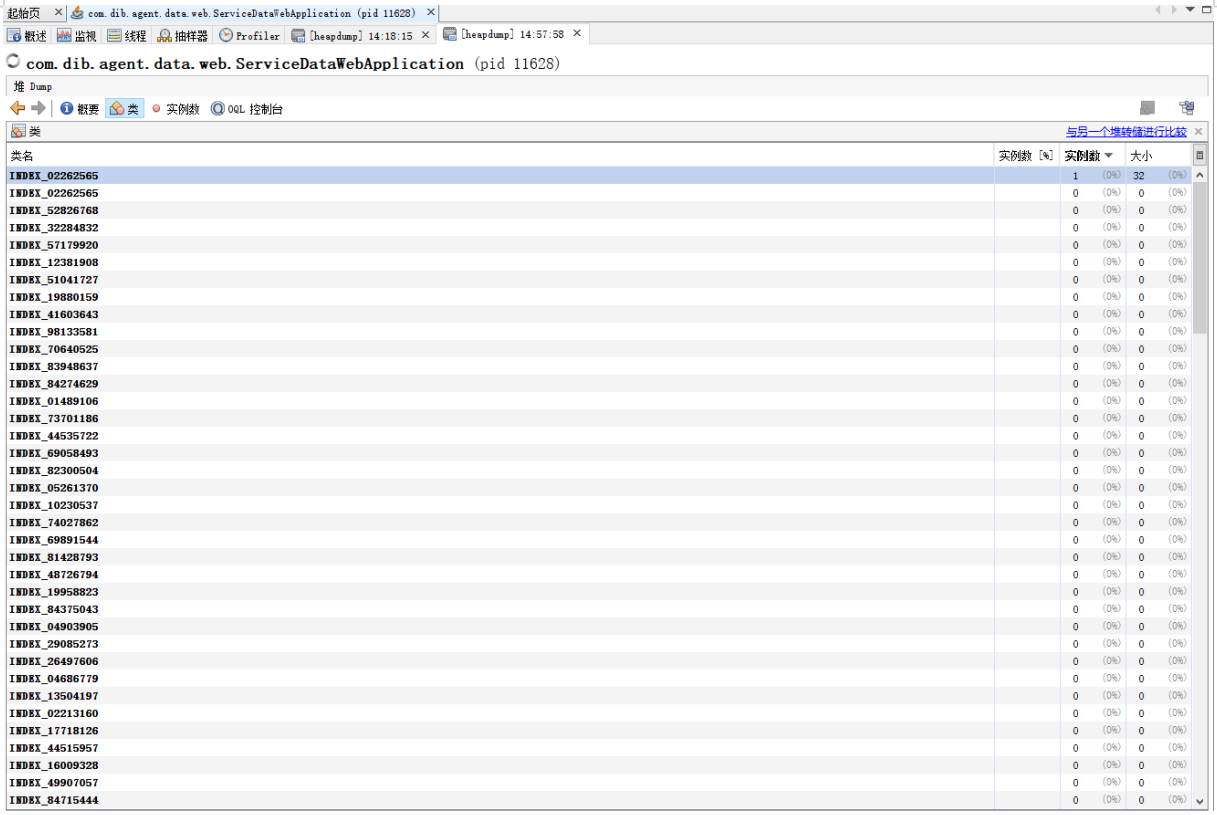
index



垃圾回收后，实例数都是0



只有刚执行的指标，存在一个实例



todo，脚本中的静态变量，估计 clazz - TB_ 的实例被垃圾回收后，仍会被保存，所以仍会存在大量的 Script_ 开头的 clz

```
c1@pingjia-VMware-Virtual-web-2:/opt/app/logs$ jstat -gcutil 1 1000 10
  S0     S1     E      O      M     CCS    YGC     YGCT    FGC    FGCT     GCT
  1.78   0.00  84.90  87.00  93.70  90.69    134    3.459     4    0.369    3.828
  1.78   0.00  84.90  87.00  93.70  90.69    134    3.459     4    0.369    3.828
  1.78   0.00  84.90  87.00  93.70  90.69    134    3.459     4    0.369    3.828
^Cc1@pingjia-VMware-Virtual-web-2:/opt/app/logs$ jcmd 1 GC.run
1:
Command executed successfully
c1@pingjia-VMware-Virtual-web-2:/opt/app/logs$ jstat -gcutil 1 1000 10
  S0     S1     E      O      M     CCS    YGC     YGCT    FGC    FGCT     GCT
  0.00   0.00   0.23   3.47  93.37  90.23    135    3.473     5    0.661    4.133
  0.00   0.00   1.41   3.47  93.37  90.23    135    3.473     5    0.661    4.133
  0.00   0.00   1.41   3.47  93.37  90.23    135    3.473     5    0.661    4.133
  0.00   0.00   1.41   3.47  93.37  90.23    135    3.473     5    0.661    4.133
  0.00   0.00   1.41   3.47  93.37  90.23    135    3.473     5    0.661    4.133
  0.00   0.00   1.41   3.47  93.37  90.23    135    3.473     5    0.661    4.133
  0.00   0.00   1.41   3.47  93.37  90.23    135    3.473     5    0.661    4.133
  0.00   0.00   1.41   3.47  93.37  90.23    135    3.473     5    0.661    4.133
  0.00   0.00   1.41   3.47  93.37  90.23    135    3.473     5    0.661    4.133
  0.00   0.00   1.41   3.47  93.37  90.23    135    3.473     5    0.661    4.133
c1@pingjia-VMware-Virtual-web-2:/opt/app/logs$
```

jcmd 1 GC.class_histogram
jmap -histo:live 1

```
        at org.apache.tomcat.util.threads.TaskThread$WrappingRunnable.run(TaskThread.java:61)
        at java.lang.Thread.run(Thread.java:750)

2025-06-18 14:59:56.968 ERROR 1 --- [http-nio-29202-exec-10] c.d.a.e.w.c.e.ComExtractRuleController   : RequestHeader account:chenlizhen
2025-06-18 14:59:56.969 ERROR 1 --- [http-nio-29202-exec-10] c.d.a.e.w.c.e.ComExtractRuleController   : account:chenlizhen
2025-06-18 15:01:42.480 ERROR 1 --- [task-1] c.d.a.e.c.r.AbstractMaterialReaderFacade : 表【ods_ss_credit_archive_s】提取失败，错误信息: 未定位到表格
2025-06-18 15:01:42.547 ERROR 1 --- [task-1] c.d.a.e.w.a.e.ComExtractDocAggregate     : 文档ID: 1379514152057352205，表名: ods_ss_credit_archive_s，提取失败，失败原因: null
2025-06-18 15:02:11.832 ERROR 1 --- [http-nio-29202-exec-1] c.d.a.e.w.c.e.ComExtractRuleController   : RequestHeader account:chenlizhen
2025-06-18 15:02:11.832 ERROR 1 --- [http-nio-29202-exec-1] c.d.a.e.w.c.e.ComExtractRuleController   : account:chenlizhen
2025-06-18 15:05:21.124 ERROR 1 --- [http-nio-29202-exec-7] c.d.a.e.w.c.e.ComExtractRuleController   : RequestHeader account:chenlizhen
2025-06-18 15:05:21.125 ERROR 1 --- [http-nio-29202-exec-7] c.d.a.e.w.c.e.ComExtractRuleController   : account:chenlizhen
2025-06-18 15:06:25.083 ERROR 1 --- [http-nio-29202-exec-10] c.d.a.e.w.c.e.ComExtractRuleController   : RequestHeader account:chenlizhen
2025-06-18 15:06:25.084 ERROR 1 --- [http-nio-29202-exec-10] c.d.a.e.w.c.e.ComExtractRuleController   : account:chenlizhen
2025-06-18 15:06:30.270 ERROR 1 --- [http-nio-29202-exec-4] c.d.a.e.w.c.e.ComExtractRuleController   : RequestHeader account:chenlizhen
2025-06-18 15:06:30.270 ERROR 1 --- [http-nio-29202-exec-4] c.d.a.e.w.c.e.ComExtractRuleController   : account:chenlizhen
2025-06-18 15:31:45.715 ERROR 1 --- [task-1] c.d.a.e.c.r.AbstractMaterialReaderFacade : 表【ods_ss_credit_archive_s】提取失败，错误信息: 未定位到表格
2025-06-18 15:31:45.762 ERROR 1 --- [task-1] c.d.a.e.w.a.e.ComExtractDocAggregate     : 文档ID: 1379514152057352205，表名: ods_ss_credit_archive_s，提取失败，失败原因: null
2025-06-18 16:01:47.588 ERROR 1 --- [task-1] c.d.a.e.c.r.AbstractMaterialReaderFacade : 表【ods_ss_credit_archive_s】提取失败，错误信息: 未定位到表格
2025-06-18 16:01:47.638 ERROR 1 --- [task-1] c.d.a.e.w.a.e.ComExtractDocAggregate     : 文档ID: 1379514152057352205，表名: ods_ss_credit_archive_s，提取失败，失败原因: null
2025-06-18 16:27:43.947 ERROR 1 --- [task-1] c.d.d.d.e.c.e.util.CellValueStandard     : 字段类型转换失败，单元格值: 5.466E-7，字段类型: DECIMAL
2025-06-18 16:27:43.989 ERROR 1 --- [task-1] c.d.d.d.e.c.e.util.CellValueStandard     : 字段类型转换失败，单元格值: 6.676E-7，字段类型: DECIMAL
2025-06-18 16:31:56.825 ERROR 1 --- [task-1] c.d.a.e.c.r.AbstractMaterialReaderFacade : 表【ods_ss_credit_archive_s】提取失败，错误信息: 未定位到表格
2025-06-18 16:31:56.871 ERROR 1 --- [task-1] c.d.a.e.w.a.e.ComExtractDocAggregate     : 文档ID: 1379514152057352205，表名: ods_ss_credit_archive_s，提取失败，失败原因: null
2025-06-18 17:02:05.146 ERROR 1 --- [task-1] c.d.a.e.c.r.AbstractMaterialReaderFacade : 表【ods_ss_credit_archive_s】提取失败，错误信息: 未定位到表格
2025-06-18 17:02:05.206 ERROR 1 --- [task-1] c.d.a.e.w.a.e.ComExtractDocAggregate     : 文档ID: 1379514152057352205，表名: ods_ss_credit_archive_s，提取失败，失败原因: null
```